THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. We decompose the Mertens function $M(x)=\sum_{n\le x}\mu(n)$ into excursions between zero runs and attach to each completed excursion a triangular skeleton. The paper has three layers, kept separate throughout: new results, new reformulations obtained by combining published theorems, and results quoted from the literature (each statement carries one of the tags [N], [N-el], [C], [L] explained in Section 1.8).

New results. (i) Elementary excursion counts combined with the Kaczorowski–Pintz oscillation theorem give horizon-localized forms of Littlewood's criterion: for every fixed $\lambda>\lambda_*:=e^{\pi/\gamma_1}$, RH is equivalent to $|M(t)|\ll_\varepsilon t^{1/2+\varepsilon}$ on $x_0<t\le\min\{x_1,\lambda x_0\}$ for the excursions $(x_0,x_1]$ of M; λ_* is the sharp threshold of the counting mechanism, and the localized window depends only on the Möbius prefix up to χu for every $\chi>\chi_*:=\lambda_*/2$. (ii) The late Mertens excursions satisfy $x_1\le x_0^A$ for every fixed $A>1$. (iii) The absolute area of the skeleton has exact exponent: $\int_0^x|M_{\mathrm{s}}|=x^{1+\Theta/2+o(1)}$, where $\Theta=\sup\Re\rho$; hence its $x^{5/4+\varepsilon}$ bound is equivalent to RH. (iv) At every zero v of M, $\sum_{n<v}M(n)^2=-\sum_{h<v}hC_h^{[v]}$ with exact cancellation of cross-excursion terms, and the signed weighted correlation is carried by lags $h>v^\alpha$ for every $\alpha<\Theta$. (v) Kaczorowski–Pintz witness excursions have mean logarithmic length at most π/γ_1; their peaks are wide; channel descent, multiplicative-layer depth and a relaxed divisor-sum class are analysed and three natural closing mechanisms are shown to fail.

Reformulations. With the published inputs of Pintz (mean and late maximal order of $|M|$), Matomäki–Radziwiłł–Tao, Matomäki–Teräväinen and Vinogradov–Korobov we obtain exact logarithmic exponents of the absolute, quadratic and chord-energy masses, a Dirichlet-block expansion of $1/\zeta$ whose abscissa of absolute block convergence is exactly Θ, uniform absolute two-point Cesàro cancellation on every interval $u^{3/5+\iota}\le L\le u$, and a Fejér reduction of a complete-excursion problem to a diagonal local-variance law.

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1. Introduction

This introduction has two parts. Sections 1.1–1.7 collect, as a short manual, the facts about the Möbius function, the Mertens function, ζ, its zeros and the Chowla-type problems that are used or referred to in the paper. Sections 1.8–1.10 describe what is proved here and classify it.

1.1. **The Möbius function and the Mertens function.** The Möbius function, introduced by Möbius [17] in the inversion formula that bears his name, is $\mu(1) = 1$, $\mu(n) = (-1)^k$ if n is a product of k distinct primes, and $\mu(n) = 0$ if $p^2 \mid n$ for some prime p. It is multiplicative and characterized by

$$\sum_{d|n} \mu(d) = \mathbf{1}_{n=1} \qquad (\mu * 1 = \delta),$$

which is the source of every recursion used below: for $n \geq 2$, $\mu(n) = -\sum_{d|n,\, d<n} \mu(d)$, and summing, $\sum_{k\leq x} M(x/k) = 1$ for $x \geq 1$ [45, §3.2]. The Mertens function $M(x) = \sum_{n\leq x} \mu(n)$ was tabulated by Mertens [16], who conjectured $|M(x)| \leq \sqrt{x}$; Stieltjes had claimed $M(x) = O(\sqrt{x})$ in 1885 [25]. Squarefree integers have density $6/\pi^2$: $\sum_{n\leq x} \mu(n)^2 = \frac{6}{\pi^2} x + O(\sqrt{x})$. Computationally, M is

evaluated by combinatorial algorithms based on the recursion above [4, 36]; Hurst [37] computed $M(x)$ for all $x \le 10^{16}$. $M(n)$ is also the determinant of the $n \times n$ Redheffer matrix [21] and, up to sign, the Euler characteristic of a cell complex constructed by Björner [1].

1.2. **Euler product, ζ and $1/\zeta$.** For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \prod_p \Big(1 - \frac{1}{p^s}\Big) = \sum_{n \ge 1} \frac{\mu(n)}{n^s} = s \int_1^\infty M(x) x^{-s-1}\, dx.$$

The prime number theorem (Hadamard [7], de la Vallée Poussin [3]) is equivalent to $M(x) = o(x)$ [12] and to $\sum_n \mu(n)/n = 0$ [30]; the Vinogradov–Korobov zero-free region gives $M(x) \ll x \exp\big(-c(\log x)^{3/5}(\log\log x)^{-1/5}\big)$ [53, 6]. We also use $\sum_n \mu(n)\log n/n = -1$ (the value at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$ up to sign, justified by the prime number theorem with error term).

1.3. **Zeros, the explicit formula and RH.** The nontrivial zeros $\rho = \beta + i\gamma$ of ζ lie in $0 < \beta < 1$, infinitely many lie on $\beta = 1/2$ (Hardy [8]), and the lowest positive ordinate is $\gamma_1 = 14.134725\ldots$. Put $\Theta = \sup \beta \in [1/2, 1]$; RH is $\Theta = 1/2$ [22], and it has been verified numerically up to height $3 \cdot 10^{12}$ [20]. If all zeros are simple and RH holds, then along a suitable sequence $T_k \to \infty$ [52, §14.27]

$$M(x) = \lim_{k \to \infty} \sum_{|\gamma| < T_k} \frac{x^\rho}{\rho \zeta'(\rho)} - 2 + \sum_{n \ge 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\, n\, \zeta(2n+1)}.$$

The size of M is governed by Θ: for every $\varepsilon > 0$,

$$M(x) \ll_\varepsilon x^{\Theta + \varepsilon}, \qquad M(x) = \Omega(x^{\Theta - \varepsilon}),$$

the upper bound by Perron's formula and the bound $1/\zeta(s) \ll |t|^\varepsilon$ to the right of $\Re s = \Theta$ (Littlewood's argument [13], [52, §14.25], [49, §4]), the lower bound because $\int_1^\infty M(x) x^{-s-1} dx$ would otherwise continue $1/(s\zeta(s))$ analytically beyond $\Re s = \Theta$. In particular Littlewood's criterion: RH holds iff $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$ [13]. Under RH the best known bound is $M(x) \ll \sqrt{x} \exp\big((\log x)^{1/2}(\log\log x)^{14}\big)$ (Soundararajan [24], improving Maier–Montgomery [14]); under RH and a hypothesis on $\sum_{\gamma \le T} |\zeta'(\rho)|^{-2}$, Ng [46] obtained a limiting distribution for $e^{-y/2} M(e^y)$. The bound $M(x) = O(\sqrt{x})$ would imply RH and simplicity of all zeros [52, Ch. XIV].

1.4. **Oscillation.** The Mertens conjecture is false (Odlyzko–te Riele [48]; effective form by Pintz [19]); currently $\limsup M(x)/\sqrt{x} > 1.826$ and $\liminf M(x)/\sqrt{x} < -1.837$ [37], so in particular $M(x) = \Omega_\pm(\sqrt{x})$ unconditionally and M changes sign, and vanishes, at arbitrarily large integers. Ingham [11] showed that $\limsup M(x)/\sqrt{x} = +\infty$ follows from RH and linear independence of the ordinates. Kaczorowski and Pintz [40] proved that the number of sign changes of M in $[1, Y]$ is at least $(\gamma_1/\pi - o(1))\log Y$, with oscillation of size $x^{\Theta - \varepsilon}$, and localized one-sided oscillation in every range $[Y^{1-\varepsilon}, Y]$ (Theorem 9.5). Pintz proved $D_M(Y) = \frac{1}{Y}\int_1^Y |M| \to \infty$ with $D_M(Y) \ge Y^{\beta_0}/(6|\gamma_0|^3)$ for every zero [18], and recently that the mean and the late maximal order of $|M|$ are both $Y^{\Theta + o(1)}$, with a finer description through the zeros [49] (Theorem 8.10).

1.5. **Liouville's function.** Liouville's function $\lambda(n) = (-1)^{\Omega(n)}$ has $\sum \lambda(n) n^{-s} = \zeta(2s)/\zeta(s)$; hence $\lambda = \mathbf{1}_\square * \mu$ and $\mu(n) = \sum_{d^2 \mid n} \mu(d)\lambda(n/d^2)$, so with $L(x) = \sum_{n \le x} \lambda(n)$,

$$L(x) = \sum_{d \le \sqrt{x}} M(x/d^2), \qquad M(x) = \sum_{d \le \sqrt{x}} \mu(d) L(x/d^2),$$

and RH is equivalent to $L(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Pólya's conjecture $L(x) \le 0$ is false [10]. Most correlation results below are proved for λ and μ by the same arguments; we state them for μ.

1.6. **Short intervals, Chowla, Cesàro and logarithmic averages.** Chowla's conjecture [2] asserts, for distinct shifts $h_1, \dots, h_k$, $\sum_{n \le x} \mu(n+h_1)\cdots\mu(n+h_k) = o(x)$ (Cesàro form); the logarithmic form replaces $\sum_{n\le x}$ by $\sum_{n\le x} \frac{1}{n}$ and $o(x)$ by $o(\log x)$, and is implied by the Cesàro form. Sarnak [23] observed that Chowla implies Möbius disjointness from zero-entropy dynamics, and Tao proved the equivalence of the logarithmic Chowla and logarithmic Sarnak conjectures [27]. The known cases are: the logarithmic two-point case (Tao [26]), the logarithmic odd-order cases (Tao–Teräväinen [28, 29]), and Cesàro statements averaged over the shifts: Matomäki–Radziwiłł–Tao [42] prove $\sum_{h\le K} |\sum_{n\le x} \mu(n)\mu(n+h)| = o(Kx)$ for every $K = K(x) \to \infty$. The two-point Cesàro case for fixed h remains unproved. These rest on the Matomäki–Radziwiłł theorem [15] on multiplicative functions in almost all short intervals. For *all* short intervals, Matomäki–Teräväinen [44] prove $\sum_{x<n\le x+x^\theta} \mu(n) \ll x^\theta (\log x)^{-1/3+\varepsilon}$ for $\theta > 0.55$ and Fourier uniformity for $H \ge x^{3/5+\varepsilon}$, extended by Matomäki–Shao–Tao–Teräväinen [43]; primes in all short intervals are due to Huxley [38] and, with the Guth–Maynard large-value estimate, to length $x^{17/30+o(1)}$ [34]. The distinction between results for all intervals and for almost all intervals is essential here, because excursion endpoints are selected by M itself.

1.7. **The walk viewpoint.** Denjoy [5] observed that RH is the statement that M grows like a random walk; Wintner [31] introduced random multiplicative functions, whose partial sums have better than square-root cancellation (Harper [9]). Areas and quadratic functionals of lattice excursions are classical [51, 47, 50, 39, 33, 32]; here they are applied to the deterministic Möbius walk.

1.8. **Classification of the results.** Every numbered statement carries one tag:

- [N] new result; it may use published theorems as input, and the input is named in the statement or proof;
- [N-el] new result with a short elementary proof;
- [C] reformulation or direct combination of published theorems, stated in excursion variables; possibly new in this form, and useful here, but its mathematical content lies in the quoted inputs;
- [L] result from the literature, quoted with reference (or a classical identity recorded for completeness).

Novelty tags record that we did not find the statement in the literature; absence of a match is weak evidence.

Entirely new. The excursion skeleton and its exact formulas; the geometric counting lemmas for disjoint intervals; the horizon-localization theorem and the descending horizons down to λx_0 with sharp counting threshold $\lambda_* = e^{\pi/\gamma_1}$ (Kaczorowski–Pintz supply only the oscillation density); the localization of returns $x_1 \le x_0^A$; the exact exponent $1 + \Theta/2$ of the skeleton area; the two-sided localization of large values; the mean logarithmic-length bound; wide peaks and packing; the prefix-dependency cutoff $\chi_* = \lambda_*/2$; channel descent and its quantified persistence; the failure of the relaxed divisor-sum class, of termwise contraction and of bounded layer depth; the weighted correlation identity at zeros and its Θ-edge lag localization; the Catalan selection energies; the large prime decomposition of M and its asymptotics $\pm t/\log^2 t$.

Useful reformulations. Littlewood's criterion in L^1, L^2, energy and correlation form; exact exponents of $B_1, B_2, \mathcal{E}$ from Pintz; the excursion-block series of $1/\zeta$; averaged Chowla, Fourier uniformity and short-interval cancellation transported to excursion blocks.

Literature. The results of Kaczorowski–Pintz, Pintz, Matomäki–Radziwiłł–Tao, Matomäki–Teräväinen, Heath-Brown's identity, the reflection principle and the classical identities for μ.

1.9. **Main results.** *Geometric counts* (Lemmas 9.1–9.4). For any ordered family of disjoint integer intervals $(x_0, x_1]$, the members starting below X with $x_1 > x_0^A$ are $O_A(\log\log X)$, with $x_1 > cx_0^2$ are $O_c(\log\log X)$, with $x_1 > x_0\psi(x_0)$ are $o(\log X)$ when $\psi \to \infty$, and with $x_1 > \lambda x_0$ are at most $\log X/\log\lambda + O(1)$.

Horizon localization (Theorem 9.11, Corollaries 9.13, 9.14). If $\Theta > 1/2$, at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting below X carry values $\geq \frac{1}{2}t^{\Theta-\eta}$. Hence any horizon whose crossing family has at most $(\kappa + o(1))\log X$ members, $\kappa < \gamma_1/\pi$, gives an RH-equivalent localization; in particular

$$\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ on } x_0 < t \leq \min\{x_1, \lambda x_0\}, \qquad \lambda > \lambda_* = e^{\pi/\gamma_1} = 1.2489\ldots$$

The erosion chain $x_0^2 \to x_0^2/2^j \to x_0\psi(x_0) \to \lambda x_0$ is rigorous for general families; for M itself the power horizons are inactive by Corollary 9.8.

Exact exponents (Corollaries 8.11, 8.13, Theorem 8.16).

$$\sum_{n\leq X}|M(n)| = X^{1+\Theta+o(1)}, \quad \sum_{n\leq X}M(n)^2 = X^{1+2\Theta+o(1)}, \quad \mathcal{E}(X) = X^{1+\Theta+o(1)}, \quad \int_0^X |M_{\mathrm{s}}| = X^{1+\Theta/2+o(1)},$$

and $\sum_I D_I(s)$ over completed excursions converges absolutely in blocks exactly for $\Re s > \Theta$, where it equals $1/\zeta(s)$.

Linear window (Section 10). For $u < t \leq \lambda u < 2u$ and $M(u) = 0$,

$$M(t) = -\sum_{d\leq \lambda u/2}\mu(d)\big(\lfloor t/d\rfloor - \lfloor u/d\rfloor\big),$$

so the RH-equivalent window is determined by the prefix up to χu for every $\chi > \chi_* = \lambda_*/2$; the channel decomposition, the kernel form and the multiplicative-layer filtration make the dependency explicit and identify where cancellation must occur.

Correlations (Section 11). At zeros v of M, $\sum_{n<v}M(n)^2 = -\sum_{h<v}hC_h^{[v]} = v^{1+2\Theta+o(1)}$, lags $h \leq \sqrt{v}$ are negligible at the v^2 scale, and lags $h \leq v^\alpha$, $\alpha < \Theta$, carry a vanishing proportion.

Extreme values (Section 12). Under a uniform tail hypothesis on a normalized excursion statistic in a stochastic model, each level of lacunarity removes one logarithm from the extreme scale.

1.10. **Organization.** Sections 3–5: excursion geometry. Sections 6–8: the relation with ζ, area bounds, global criteria, exact exponents and the block series of $1/\zeta$. Section 9: counts, Kaczorowski–Pintz input, horizon localization, Catalan reference laws, wide peaks. Section 10: the localized window and the large-prime decomposition. Section 11: correlations. Section 12: stochastic models. Then a one-frequency model, numerical data, and conclusions.

Conventions. $\rho = \beta + i\gamma$ runs over nontrivial zeros; $\Theta = \sup\beta$; $\gamma_1 = 14.134725\ldots$; RH means $\Theta = 1/2$. $M(x) = M(\lfloor x\rfloor)$ for real x, $M(x) = 0$ for $x < 1$. Heuristics are marked as such.

2. Notation and variables

Global symbols keep the meanings below. Symbols marked "local" are declared again where used and never leave that statement or subsection. Constants c, C are positive and may change from line to line.

Symbol	Meaning	Scope
$\rho = \beta + i\gamma$	Nontrivial zero of ζ; β is used only here	Global
Θ	$\sup\{\Re\rho\}$	Global
γ_1	Lowest positive ordinate, $14.134725\ldots$	Global
$\lambda_*,\ \chi_*$	e^{π/γ_1}, $\lambda_*/2$	Global
$M,\ M_0$	Mertens function; $M_0 = M + 2$ (normalization of [40])	Global
$I_\nu = (x_{0,\nu}, x_{1,\nu}]$	Completed excursion; also $(u_\nu, v_\nu]$	Global
σ_ν	Orientation ±1 of I_ν	Global
$\ell_\nu,\ H_\nu,\ \tau_\nu$	Length, height, first apex	Global
$a_\nu,\ b_\nu$	$\tau_\nu - x_{0,\nu}$, $x_{1,\nu} - \tau_\nu$	Global
$\mathcal{Z} = [p, q]$	Zero run	Section 4
$u(x),\ L_x$	Last zero $\leq x$; terminal length $\lfloor x\rfloor - u(x)$	Global

Symbol	Meaning	Scope
$M_{\mathrm{s}},\ M_{\mathrm{sk}}$	Regularized / unregularized skeleton	Global
$\mathcal{A}_+,\ \mathcal{A}_-,\ \mathcal{A},\ \mathcal{A}_{\mathrm{abs}}$	Positive, signed negative, net, absolute area	Global
$\widetilde{A}$	$\int_0^x M$	Section 6
ch_ν	Chord of $\mathcal{A}$ over I_ν	Theorem 8.9
$\mathfrak{q}_\nu = q_\nu$	Number of active steps $\#\{n \in I_\nu : \mu(n) \neq 0\}$; q in a general block	Global
$Q_\nu,\ \mathfrak{b}_\nu$	$\sum_{n\in I_\nu} M(n)^2$, $\sum_{n\in I_\nu} \lvert M(n)\rvert$	Global
$Q_{\mathrm{term}},\ \mathfrak{b}_{\mathrm{term}}$	Same for the terminal open prefix	Global
$B_1,\ B_2,\ \mathcal{E}$	$\sum_{n\le X} \lvert M\rvert$, $\sum_{n\le X} M^2$, chord energy	Global
$C_h^{(\nu)},\ C_h^{[v]},\ C_h^{(u,X]}$	Lag-h correlation in I_ν, in $(0,v]$, in $(u,X]$	Global
$C_g^{\langle h\rangle}$	Lag-g correlation in a truncated prefix of length h	Local
S_t	Oriented partial sum inside a declared block	Local
$\mathcal{F},\ E_{\mathcal{F}}(X)$	Family of excursions and its counting function	Section 9
$\mathfrak{h},\ \mathcal{F}_{\mathfrak{h}}$	Horizon and crossing family	Section 9
A	Power exponent in $x_1 > x_0^A$	Section 9
η	Loss in the Kaczorowski–Pintz exponent $\Theta - \eta$	Global
κ	Coefficient of a crossing count $(\kappa + o(1))\log X$	Theorem 9.11
$\mathcal{S},\ L^{\#},\ r_I$	Squarefree clock, compressed length, support time	Catalan subsection
L	Length of a declared block (also support length $L^{\#}$ in Catalan lemmas)	Local
$p_{\mathrm{Cat}},\ p_{\mathrm{Cat}}^{\max},\ p_{\mathrm{open}}$	Reference-law tail weights	Catalan subsection
$\theta_{\mathrm{w}},\ \theta'$	Short-interval exponents > 0.55	Wide peaks
$\Lambda,\ h_u,\ \omega_u$	Horizon factor, block length, Lipschitz modulus	Wide peaks
G	$(H - h_u)_+$	Proposition 9.32
μ_u	$\mu\mathbf{1}_{[1,u]}$	Section 10
$\Delta_k,\ \mathcal{R}_K$	Channel increment, deep remainder	Section 10
ϑ	Normalizing exponent in $\lvert M(y)\rvert/y^{\vartheta}$	Section 10
$c_y(t,u),\ K_y$	Kernel coefficients (argument y of $M(y)$)	Section 10
$\Upsilon,\ \mathcal{C}$	Multiplicity-weighted and coherent coefficients	Section 10
$\mathbf{a} = (a_d),\ D_{\mathbf{a}}$	Relaxed prefix vector and its divisor sum	Section 10
P	Prefix cutoff in layer statements	Section 10
$P^+(n)$	Largest prime factor	Section 10
$U,\ W,\ M_U^{\mathrm{sm}},\ M_U^{\mathrm{lp}},\ \widetilde{m}$	Large-prime level, t/U^2, smooth and large-prime parts, $\sum_{a\le y}\mu(a)/a$	Section 10
$\Omega(n),\ \omega(n)$	Prime factors with / without multiplicity	Global
$D_I(s)$	Dirichlet block of a completed excursion	Section 8
$s = \varsigma + i\tau$	Complex variable	Global
$e(x),\ \xi$	$e^{2\pi i x}$; frequency	Section 11
K	Lag cutoff in averaged-Chowla statements	Section 11
$R,\ F_R,\ V_R$	Fejér scale and sums	Section 11
ι	Exponent gap in $L \ge u^{3/5+\iota}$	Section 11
$(\mathfrak{X}, \mathcal{G}, \Pr),\ Z_k$	Probability space and excursion statistic	Section 12
$\mathcal{N}(X)$	Deterministic count bound	Section 12
$\kappa_0,\ \varpi$	Tail constant and exponent	Section 12

3. Excursions and the regularized skeleton

Let μ be the Möbius function and

$$M(x) = \sum_{n\le x} \mu(n), \qquad x \ge 0, \tag{3.1}$$

piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$. Throughout the paper, for real $x \geq 0$ we use the convention

$$M(x) = M(\lfloor x \rfloor),$$

and we set $M(x) = 0$ for $x < 1$. Thus expressions such as $M(t/k)$ always mean $M(\lfloor t/k \rfloor)$. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [37] computed $M(x)$ for all $x \leq 10^{16}$. The area and quadratic functionals of lattice excursions are classical [51, 47, 50, 39, 33, 32]; here they are applied to the deterministic Möbius walk.

Definition 3.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}]$$

with integer endpoints, $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu} + 1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \geq 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$. Consecutive excursions may be separated by a *zero run* $[x_{1,\nu}, x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. The first excursion is $(0, 2]$. Since M assumes positive and negative values at arbitrarily large arguments ($M(x) = \Omega_\pm(\sqrt{x})$, Remark 7.6) and its integer jumps belong to $\{-1, 0, 1\}$, between two integers where M has opposite signs it must vanish at an integer. Hence M has arbitrarily large zeros and every excursion is finite. We also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu} < n < x_{1,\nu}} |M(n)|,$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu.$$

Every excursion has $H_\nu \geq 1$ and $\ell_\nu \geq 2$.

Definition 3.2 (Completed excursions and open prefixes)**.** An excursion in the sense of Definition 3.1 is always *completed*: both $M(x_{0,\nu}) = 0$ and $M(x_{1,\nu}) = 0$ are part of the definition. A segment $(u, x]$ with $M(u) = 0$ and $M(n) \neq 0$ for $u < n \leq x$, with no return condition at x, is an *open prefix*; the terminal block of Definition 3.6 is one. Any estimate whose proof uses the backward bound $|M(u + r)| \leq x_{1,\nu} - u - r$ holds for completed excursions only and is never applied to an open prefix.

Lemma 3.3 (Height and step count)**.** [N-el] *For every excursion,*

$$H_\nu \leq \min(a_\nu, b_\nu) \leq \frac{\ell_\nu}{2}. \tag{3.2}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 3.4 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 3.5 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu} \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \leq t \leq \tau_\nu, \\ \sigma_\nu \sqrt{H_\nu} \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \leq t \leq x_{1,\nu}, \end{cases} \tag{3.3}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. Here "regularized" refers only to the normalization of the apex from H_ν to $H_\nu^{1/2}$; no analytic smoothing operation is intended. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 3.6 (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since M has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u<n\leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}_{\mathrm{term}}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big].$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real x with last zero $u(x) \leq \lfloor x \rfloor$ define

$$\mathfrak{b}_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x \rfloor - u(x).$$

These depend only on M.

Definition 3.7 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\,dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\,dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\,dt. \tag{3.4}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign. The curve $\mathcal{A}_-$ records negative area with its sign; it is not the absolute value of the negative area.

Remark 3.8 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0, \infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \geq 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 5 concern the Möbius increments and do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization.

4. Exact formulas

Lemma 4.1 (Increment per excursion)**.** [N-el] *For every complete excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu\sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{4.1}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. For the discrete sum, the ascending values at $j = 1, \ldots, a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\leq a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu + 1)/2$; the descending values at $\tau_\nu + r$, $r = 1, \ldots, b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\leq b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu - 1)/2$. □

Proposition 4.2 (Explicit form of the three curves)**.** [N-el] *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x - x_{0,\nu})^2}{2a_\nu}, & x \leq \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x - \tau_\nu) - \dfrac{(x - \tau_\nu)^2}{2b_\nu}\right), & x \geq \tau_\nu. \end{cases} \tag{4.2}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\leq k} \sigma_\nu\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\leq k\\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\leq k\\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{4.3}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (3.3); the endpoint formulas follow from Lemma 4.1. □

A point y is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of y and strictly concave on the other, within distance δ. The second derivative need not exist at y.

Lemma 4.3 (Critical geometry of $\mathcal{A}$)**.** [N-el] *Let $\mathcal{Z} = [p, q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then $\mathcal{Z}$ is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_-$, then $\mathcal{A}$ has no local extremum on $\mathcal{Z}$. Its concavity has sign $-\sigma_-$ immediately to the left and $+\sigma_+$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of $\mathcal{Z}$ is a strict inflection, and the change of concavity is carried by the whole plateau.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. Inside an excursion $\mathcal{A}' = M_{\mathrm{s}} \neq 0$, so no extremum occurs there. On $\mathcal{Z}$, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

5. Internal correlations of an excursion

For a real block $x_1, \dots, x_L$ put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{5.1}$$

Lemma 5.1 (Triangular identity)**.** [N-el] *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} h C_h = S_L \sum_{t=1}^{L-1} S_t. \tag{5.2}$$

Proof. Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j \big(L + 1 - \max(i, j)\big).$$

Write $\max(i,j) = \frac{i+j+|i-j|}{2}$. Since $\frac{1}{2}\sum_{i,j}(i+j)x_i x_j = S_L \sum_i i x_i$ and each unordered pair at lag $h = |i-j| \geq 1$ occurs twice,

$$\sum_{t=1}^{L} S_t^2 = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \frac{1}{2}\sum_{i,j=1}^{L} x_i x_j |i-j| = (L+1)S_L^2 - S_L \sum_{i=1}^{L} i x_i - \sum_{h=1}^{L-1} h C_h.$$

Moreover $\sum_i i x_i = L S_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h h C_h$; subtract S_L^2. □

For a bridge, $S_L = 0$, so the preceding identity gives

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} h C_h. \tag{5.3}$$

Remark 5.2 (Hermitian form). For complex x_j and $C_h = \sum_{j=1}^{L-h} x_j\overline{x_{j+h}}$, the same computation applies. The ordered pairs (i,j) and (j,i) at lag h contribute $x_i\overline{x_j} + x_j\overline{x_i}$, so each lag contributes $2\Re C_h$ to $\sum_{i,j} x_i\overline{x_j}|i-j|$, and the cross term becomes $\Re(\overline{S_L}\sum_i ix_i)$. Hence

$$\sum_{t=1}^{L-1}|S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L}\sum_{t=1}^{L-1} S_t\right).$$

For a complete excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu}+j), \qquad S_t = |M(x_{0,\nu}+t)|. \tag{5.4}$$

Then $S_t > 0$ for $1 \le t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu-1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu-h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu-1} hC_h^{(\nu)}. \tag{5.5}$$

Lemma 5.3 (Zeroth moment and extreme lag)**.** [N-el] *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu-1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu-1}^{(\nu)} = -1, \tag{5.6}$$

and q_ν is even and at least 2.

Proof. $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 5.4 (Quadratic decomposition by excursions)**.** [N-el] *For real $x \ge 1$ let $u(x)$ and L_x be as in Definition 3.6, and for $1 \le j \le L_x$ put $y_j = \mu(u(x)+j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x-h} y_j y_{j+h}$. Then*

$$\sum_{n\le x} M(n)^2 = \sum_{\nu: x_{1,\nu}\le u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{5.7}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x\rfloor)^2 + M(\lfloor x\rfloor)\sum_{t=1}^{L_x-1} M(u(x)+t) - \sum_{h=1}^{L_x-1} hC_h^{\mathrm{term}}, \tag{5.8}$$

with $Q_{\mathrm{term}}(x) = 0$ if $L_x = 0$. At a zero x of M,

$$\sum_{n\le x} M(n)^2 = \sum_{x_{1,\nu}\le x} Q_\nu = -\sum_{x_{1,\nu}\le x}\sum_h hC_h^{(\nu)}. \tag{5.9}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 5.1 to the terminal block and add its final term $S_{L_x}^2$. □

Lemma 5.5 (Elementary bound for an open terminal block)**.** [N-el] *Let u be a zero of M and $L \ge 0$. Then $|M(u+r)| \le r$ for $1 \le r \le L$, and*

$$\sum_{r=1}^{L} M(u+r)^2 \le \frac{L(L+1)(2L+1)}{6}. \tag{5.10}$$

In particular $Q_{\mathrm{term}}(x) \ll L_x^3$. The bridge bound (5.17)*, whose leading coefficient is* $1/12$ *rather than* $1/3$*, applies only to complete excursions.*

Proof. $M(u+r) = \sum_{j\le r}\mu(u+j)$ with $|\mu| \le 1$; square and sum. □

Thus the quadratic sum sees only lag correlations internal to excursions: pairs of Möbius increments in different completed excursions do not contribute.

Let

$$\mathfrak{b}_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t = \sum_{n \in I_\nu} |M(n)|. \tag{5.11}$$

Cauchy–Schwarz gives

$$\mathfrak{b}_\nu^2 \le (\ell_\nu - 1) Q_\nu, \tag{5.12}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu - 1} h\big(2D_h^{(\nu)} - P_h^{(\nu)}\big). \tag{5.13}$$

Proposition 5.6 (Sharp bounds at fixed length and height)**.** [N-el] *Let $S_0 = S_L = 0$, $S_t \ge 1$ for $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ and $H = \max S_t$. Then $L \ge 2H$ and*

$$H^2 + L - 2H \le \mathfrak{b} \le H(L - H), \tag{5.14}$$

$$\frac{2H^3 + H}{3} + L - 2H \le Q \le \frac{2H^3 + H}{3} + (L - 2H)H^2. \tag{5.15}$$

All four bounds are attained. In particular

$$Q \ge \frac{2H^3 + H}{3}, \qquad Q \ge L - 1. \tag{5.16}$$

Proof. Before and after the first visit to H the path visits each level $1, \dots, H-1$. Together with one visit to H, these $2H - 1$ designated interior times contribute $2(1 + \cdots + (H-1)) + H = H^2$ to $\mathfrak{b}$ and $2\sum_{j<H} j^2 + H^2 = (2H^3 + H)/3$ to Q. The remaining $L - 2H$ interior times lie at heights between 1 and H, which gives the four inequalities. The lower bounds are attained by inserting the $L - 2H$ remaining steps as horizontal steps at height 1 into the tent path of height H; the upper bounds by inserting all of them as horizontal steps at height H. Both constructions keep $S_t \ge 1$ on $0 < t < L$ and have increments in $\{-1, 0, 1\}$. □

Corollary 5.7 (Sharp fixed-length bounds)**.** [N-el] *For $L \ge 2$,*

$$\mathfrak{b} \le \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \le \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases}$$

and in particular

$$Q \le \frac{L^3 + 2L}{12}. \tag{5.17}$$

Both bounds are attained by the tent path, with one horizontal step at maximal height when L is odd.

Proof. Use $S_t \le \min(t, L - t)$ and sum this bound and its square. □

6. Where ζ enters

The black curve depends nonlinearly on M. The dependence enters through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$; no closed expression of its Mellin transform in terms of ζ is known. The

nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of M. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer x,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\ge 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{6.1}$$

[52, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has $\widetilde{A}(1) = 0$ (the value of M at the single point 1 does not affect the integral). Integration by parts therefore gives, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{6.2}$$

The residues of $x^{s+1}/(s(s+1)\zeta(s))$ are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{6.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s = 0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s = -1$. Under RH and simplicity of the zeros, the contour argument of [52, §14.27], with the extra decay $(s+1)^{-1}$, turns (6.3) into an identity for $\widetilde{A}(x)$ with the ρ-sum taken as $\lim_k \sum_{|\gamma|<T_k}$; absolute convergence of the ρ-sum needs moment bounds for $1/\zeta'(\rho)$ of the type assumed in [46]. Formula (6.3) is not used in any proof. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms.

Proposition 6.1 (Dipole identity)**.** [N-el] *For each excursion let its* legs *be the integer sets* $J = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ *and* $(\tau_\nu, v_\nu] \cap \mathbb{Z}$*, with midpoints* $c_J = (u_\nu + \tau_\nu)/2$ *and* $(\tau_\nu + v_\nu)/2$*. Then for every zero* v_k*,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{J\subset(0,v_k]} \sum_{m\in J} \mu(m)\,(m - c_J). \tag{6.4}$$

Proof. Since M is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma_\nu H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma_\nu H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_J)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma_\nu H b/2$ and $\int_\tau^v M = b\,\sigma_\nu H + \sum_{m=\tau+1}^{v} \mu(m)(v - m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma_\nu H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_J$. On zero runs both integrands vanish. Sum over legs. □

7. Bounds for the area curves

Lemma 7.1 (Transfer)**.** [N-el] *Let* $x \ge 1$ *and let* $x^* \ge x$ *be a zero of* M *(in a computation truncated at* $N \ge x$*, take* $x^* = N$ *and the terminal skeleton of Definition 3.6). If* $|M(n)| \le C_0\, n^{1/2}$ *for all* $1 \le n \le x^*$ *(so necessarily* $C_0 \ge 1$*), then*

$$|M_{\mathrm{s}}(t)| \le C_0^{1/2} t^{1/4} \quad (0 \le t \le x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \le \tfrac{4}{5}\, C_0^{1/2} x^{5/4}. \tag{7.1}$$

Proof. For $t \ge \tau_\nu$, $|M_{\mathrm{s}}(t)| \le \sqrt{H_\nu} \le C_0^{1/2}\tau_\nu^{1/4} \le C_0^{1/2} t^{1/4}$. For $u_\nu \le t \le \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $C_0^{1/2} t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $C_0^{1/2} u_\nu^{1/4} \ge 0$ and $C_0^{1/2}\tau_\nu^{1/4} \ge \sqrt{H_\nu}$ (as $\tau_\nu \le x^*$) are at least as high; that chord lies below $C_0^{1/2} t^{1/4}$ by concavity. Integrate. □

Lemma 7.2 (A concave majorant)**.** [N-el] *Let* $\phi(y) = (\log y)^{3/5}(\log\log y)^{-1/5}$ *and, for* $\kappa > 0$*,* $F(y) = y^{1/2} e^{-\kappa\phi(y)}$*. There is* $y_0 = y_0(\kappa)$ *such that* F *is increasing and concave on* $[y_0, \infty)$*.*

Proof. With $y = e^s$ and $g(s) = s^{3/5}(\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \frac{1}{2} - \kappa g'(s)$. Then

$$yF'(y) = F\,h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big).$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \frac{1}{2}$ and $h^2 - h - \kappa g'' \to -\frac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large y. □

Corollary 7.3. [C] *From the Vinogradov–Korobov bound* [53]

$$M(x) \ll x \exp\bigl(-c\,\phi(x)\bigr) \tag{7.2}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\bigl(-\tfrac{c}{2}\,\phi(x)\bigr). \tag{7.3}$$

Proof. Let K be such that $|M(y)| \le K y\, e^{-c\phi(y)}$ for $y \ge y_1$, and let F be as in Lemma 7.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \le K^{1/2} F(y)$ for $y \ge y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \ge Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \le K^{1/2} F(\tau_\nu)$; the bound is taken at the apex τ_ν, which lies in the excursion, and not at its future endpoint v_ν. For $t \ge \tau_\nu$ monotonicity of F gives $|M_{\mathrm{s}}(t)| \le \sqrt{H_\nu} \le K^{1/2} F(t)$. For $u_\nu \le t \le \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2}F(u_\nu))$ to $(\tau_\nu, K^{1/2}F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2}F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \le K^{1/2} F(t)$ for all $t \ge Y_0$ outside the finitely many early excursions, and since F is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \le O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \le O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \quad \square$$

Corollary 7.4. [C] *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{7.4}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false. By Corollary 8.13 the converse also holds: (7.4) *is equivalent to RH.*

Proof. Under RH, $|M(y)| \le C_\varepsilon y^{1/2+\varepsilon}$, so $\sqrt{H_\nu} \le C_\varepsilon^{1/2} \tau_\nu^{1/4+\varepsilon/2}$. The chord argument of Lemma 7.1, with the concave majorant $C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ in place of $C_0^{1/2} t^{1/4}$, gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{1/4+\varepsilon/2}$ and, after integration, $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon/2}$. □

Remark 7.5 (Endpoint-indexed bounds). From $\ell_\nu \ge 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$, indexed by the future endpoint v_ν. For a general family of disjoint intervals this gives no bound at interior points. For M itself Corollary 9.8 gives $v_\nu \le u_\nu^{1+\varepsilon}$ for all late excursions, so the bound becomes $|M(t)| \ll t^{5/6+O(\varepsilon)}$; the sharp statement is the exact area exponent of Corollary 8.13.

Remark 7.6 (Oscillation of the heights). The Mertens conjecture is false [48, 41]; the current bounds are due to Hurst [37], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point n with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by v_ν does not follow, since v_ν may be much larger than n.

8. Global criteria equivalent to RH

The Mellin identity and the global L^1/L^2 equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas rewrite them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x) x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{8.1}$$

If $\int_1^\infty |M(x)| x^{-\varsigma-1} dx < \infty$ for every $\varsigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so ζ has no zeros there.

Proposition 8.1 (L^1 criterion)**.** [C] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \mathfrak{b}_\nu + \mathfrak{b}_{\mathrm{term}}(x) = \sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{8.2}$$

Proof. RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Conversely, for $\varsigma > 1/2$ choose $0 < \varepsilon < \varsigma - 1/2$; partial summation gives $\sum_n |M(n)| n^{-\varsigma-1} < \infty$, and since M is constant on $[n, n+1)$ the remark after (8.1) applies. □

Lemma 8.2 (Pruning)**.** [N-el] $\mathfrak{b}_\nu \le H_\nu(\ell_\nu - 1) \le \ell_\nu^2/2$, *hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \mathfrak{b}_\nu \le \frac{x L_0}{2}. \tag{8.3}$$

Corollary 8.3 (Reduction to long excursions)**.** [C] *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \le x \\ \ell_\nu > L_0}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_x > L_0\}} \mathfrak{b}_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{8.4}$$

At most x/L_0 long excursions are involved.

Proof. If RH holds, the left side is at most $\sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}$. Conversely, the complete excursions with $\ell_\nu \le L_0$ contribute at most $xL_0/2 \le x^{3/2}$ by Lemma 8.2, and if $L_x \le L_0$ the terminal block contributes $\mathfrak{b}_{\mathrm{term}}(x) \le \sum_{r \le L_x} r = L_x(L_x+1)/2 \le L_0^2 \le x$. Hence $\sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}$, and Proposition 8.1 applies. □

Remark 8.4 (The $\sqrt{x}$ frontier). The cut may depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\le x^{3/2+\varepsilon/2}/2$. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\mathfrak{b}_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

Heuristic 8.5 (Renewal tail)**.** If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{8.5}$$

and with $\mathfrak{b}_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (8.2), so the terminal scales $L \asymp x$ dominate the L^1 sum.

Theorem 8.6 (Correlation criterion)**.** [C] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n \le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x} \sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{8.6}$$

Proof. Under RH, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n \le x} |M(n)| \le x^{1/2} \Big(\sum_{n \le x} M(n)^2 \Big)^{1/2} \ll x^{3/2+\varepsilon/2};$$

apply Proposition 8.1. □

Corollary 8.7 (Local sufficient conditions)**.** [C] *Suppose that for every $\varepsilon > 0$ there is C_ε such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \text{for every complete excursion with } \ell_\nu > v_\nu^{1/2}. \tag{8.7}$$

Then RH holds.

Proof. Let $\varepsilon_0 \in (0,1)$, put $\varepsilon = \varepsilon_0/4$, and let C_ε be as in the hypothesis. Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (5.17) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon (\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a complete excursion, and (5.17) does not apply to it. By Lemma 5.5, $Q_{\mathrm{term}}(x) \le L_x(L_x+1)(2L_x+1)/6$, which is $O(x^{3/2})$ if $L_x \le x^{1/2}$. If $L_x > x^{1/2}$, then $M(\lfloor x \rfloor) \ne 0$. Let $I_\nu = (u(x), v_\nu]$ be the unique completed excursion whose open prefix contains $\lfloor x \rfloor$; since all summands are nonnegative, $Q_{\mathrm{term}}(x) \le Q_\nu$. Fix $A = 1+\varepsilon$. By Corollary 9.8, $v_\nu \le u(x)^A \le x^A$ once x is large, so $\ell_\nu \le x^A$. If $\ell_\nu > v_\nu^{1/2}$ the hypothesis gives $Q_\nu \le C_\varepsilon x^{A(2+\varepsilon)}$; otherwise (5.17) gives $Q_\nu \ll \ell_\nu^3 \le v_\nu^{3/2} \le x^{3A/2}$. In all cases $Q_{\mathrm{term}}(x) \ll_\varepsilon x^{(1+\varepsilon)(2+\varepsilon)} \le x^{2+4\varepsilon} = x^{2+\varepsilon_0}$. This step uses the arithmetic input of Corollary 9.8: for a general family of disjoint intervals, the future return of the terminal block is not controlled by x, and a bound for complete excursions would not control $Q_{\mathrm{term}}(x)$. Altogether $\sum_{n\le x} M(n)^2 \ll_{\varepsilon_0} x^{2+\varepsilon_0}$ for every $\varepsilon_0 \in (0,1)$, and Theorem 8.6 gives RH. □

Heuristic 8.8 (Ballistic excursions)**.** In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded k, then by (5.16) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version of (8.7) fails for every $\varepsilon \in (0,1)$. Such excursions have $\ell_\nu \ll v_\nu^{1/2}$ and do not affect the pruned condition (8.7); this is why the pruning in Corollary 8.7 is needed.

Theorem 8.9 (Energy criterion)**.** [C] *With u the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x} |M(n)|$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u) H_{\mathrm{run}}(x) \Big). \tag{8.8}$$

For a complete excursion let ch_ν be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By (4.1), $\mathrm{ch}'_\nu = \sigma_\nu \sqrt{H_\nu}/2$, so $\frac{1}{4} H_\nu \ell_\nu = \int_{I_\nu} (\mathrm{ch}'_\nu)^2$ is the Dirichlet energy of the chord. Then

$$\sum_{n\le x} |M(n)| \le 4\mathcal{E}(x) \le x \max_{n\le x} |M(n)|, \tag{8.9}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\mathfrak{b}_\nu \le H_\nu \ell_\nu$, $\mathfrak{b}_{\mathrm{term}}(x) \le (x-u) H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 8.1. □

8.1. **Exact logarithmic exponents from Pintz's mean and late-maximal-order theorem.** The criteria above are upper-bound equivalences. Theorems 2.1 and 2.2 of Pintz [49, Theorems 2.1–2.2, equations (2.14)–(2.15)] determine, in logarithmic scale, the mean modulus and late maximal order of $|M|$ in terms of the rightmost zeros of ζ. For $\Theta < 1$ the logarithmic-scale lower bound for D_M already follows from Pintz's inequality $D_M(Y) \ge Y^{\beta_0}/(6|\gamma_0|^3)$ for $Y > e^{|\gamma_0|+4}$, valid for every zero $\beta_0 + i\gamma_0$ [18], and the upper bound from the classical estimate $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$; the case $\Theta = 1$ is the new content of [49]. We record the consequence for the excursion quantities used in this paper.

Theorem 8.10 (Pintz, logarithmic mean and late maximal order [49, Theorems 2.1 and 2.2])**.** [L] *Let $\Theta = \sup\{\Re\rho : \zeta(\rho) = 0\}$. Then*

$$D_M(X) := \frac{1}{X} \int_1^X |M(t)|\, dt = X^{\Theta + o(1)}.$$

Moreover, for every fixed $\delta > 0$,

$$S_{M,\delta}(X) := \max_{X^{1-\delta} \le t \le X} |M(t)| = X^{\Theta + o_\delta(1)}.$$

Here the equalities are in logarithmic scale, equivalently

$$\frac{\log D_M(X)}{\log X} \to \Theta, \qquad \frac{\log S_{M,\delta}(X)}{\log X} \to \Theta \quad \textit{for each fixed } \delta > 0.$$

Corollary 8.11 (Exact exponents of absolute Mertens mass, quadratic mass, and chord energy)**.** [C] *Put*

$$B_1(X) := \sum_{n \le X} |M(n)|, \qquad B_2(X) := \sum_{n \le X} M(n)^2.$$

Then

$$B_1(X) = X^{1+\Theta+o(1)}, \qquad B_2(X) = X^{1+2\Theta+o(1)}, \qquad \mathcal{E}(X) = X^{1+\Theta+o(1)}. \tag{8.10}$$

Consequently

$$\Theta = \lim_{X\to\infty} \left(\frac{\log B_1(X)}{\log X} - 1 \right) = \frac{1}{2} \lim_{X\to\infty} \left(\frac{\log B_2(X)}{\log X} - 1 \right) = \lim_{X\to\infty} \left(\frac{\log \mathcal{E}(X)}{\log X} - 1 \right). \tag{8.11}$$

In particular RH is equivalent to any one of the three logarithmic exponents being respectively $3/2$*,* 2*, and* $3/2$*.*

Proof. Let $N = \lfloor X \rfloor$. Since $M(t) = M(n)$ on $[n, n+1)$,

$$\int_1^X |M(t)|\, dt = \sum_{n=1}^{N-1} |M(n)| + (X - N)|M(N)|,$$

so

$$B_1(X) = \int_1^X |M(t)|\, dt + O(|M(N)| + 1).$$

The boundary term is $O(X)$, whereas Theorem 8.10 gives $\int_1^X |M(t)|\, dt = X^{1+\Theta+o(1)}$ and $\Theta \ge 1/2$; hence $B_1(X) = X^{1+\Theta+o(1)}$. We first note that

$$\max_{n \le X} |M(n)| = X^{\Theta+o(1)}. \tag{8.12}$$

The lower bound follows from $S_{M,\delta}$ with any fixed δ. For the upper bound, if $\Theta < 1$, fix once and for all $\delta \in (1-\Theta, 1)$, independent of X; the term $o_\delta(1)$ of Theorem 8.10 is then used for this fixed δ and split at $X^{1-\delta}$: the early range is trivially at most $X^{1-\delta}$ and the late range is controlled by Theorem 8.10; when $\Theta = 1$ the trivial bound $|M(n)| \le n$ suffices.

Cauchy–Schwarz and (8.12) now give

$$\frac{B_1(X)^2}{X} \le B_2(X) \le X \max_{n \le X} |M(n)|^2,$$

so both sides have exponent $1 + 2\Theta$. Finally (8.9) squeezes $4\mathcal{E}(X)$ between $B_1(X)$ and

$$X \max_{n \le X} |M(n)|,$$

both of exponent $1 + \Theta$. □

Corollary 8.12 (Length localization of the area and quadratic mass)**.** [C] *Fix* $0 \le \alpha < \Theta$*. Among the complete excursions ending at most* X*, the total contribution of those with* $\ell_\nu \le X^\alpha$ *is negligible:*

$$\sum_{\substack{v_\nu \le X \\ \ell_\nu \le X^\alpha}} \mathfrak{b}_\nu + \mathbf{1}_{\{L_X \le X^\alpha\}} \mathfrak{b}_{\mathrm{term}}(X) = o(B_1(X)),$$

and

$$\sum_{\substack{v_\nu \le X \\ \ell_\nu \le X^\alpha}} Q_\nu + \mathbf{1}_{\{L_X \le X^\alpha\}} Q_{\mathrm{term}}(X) = o(B_2(X)).$$

Thus, on the logarithmic scale forced by the zeta zeros, essentially all L^1 and L^2 mass is carried by excursion blocks longer than X^α for every $\alpha < \Theta$.

Proof. Lemma 8.2 bounds the first complete-excursion sum by $O(X^{1+\alpha})$, while the short terminal block contributes $O(X^{2\alpha})$. Both are $o(X^{1+\Theta+o(1)})$ when $\alpha < \Theta$. For the quadratic mass use (5.17) and disjointness:

$$\sum_{\ell_\nu \le X^\alpha} Q_\nu \ll X^{1+2\alpha}, \qquad Q_{\mathrm{term}}(X) \ll X^{3\alpha} \quad (L_X \le X^\alpha).$$

Because $\alpha < \Theta \le 1$, both exponents are strictly below $1 + 2\Theta$. □

Corollary 8.13 (Exact exponent of the skeleton area)**.** [N] *Put $\mathcal{A}_{\mathrm{abs}}(x) := \mathcal{A}_+(x) + |\mathcal{A}_-(x)| = \int_0^x |M_{\mathrm{s}}(t)|\,dt$. Then*

$$\mathcal{A}_{\mathrm{abs}}(x) = x^{1+\Theta/2+o(1)} \qquad (x \to \infty).$$

In particular RH $\iff \mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{5/4+\varepsilon}$ *for every $\varepsilon > 0$.*

Proof. Upper bound. By (8.12), for every $\varepsilon > 0$ there is C_ε with $H_\nu = |M(\tau_\nu)| \le C_\varepsilon \tau_\nu^{\Theta+\varepsilon}$. Since $\Theta+\varepsilon < 2$, the function $C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$ is concave and increasing, and the chord argument of Lemma 7.1 gives $|M_{\mathrm{s}}(t)| \le C_\varepsilon^{1/2} t^{(\Theta+\varepsilon)/2}$, hence $\mathcal{A}_{\mathrm{abs}}(x) \ll_\varepsilon x^{1+(\Theta+\varepsilon)/2}$.

Lower bound. Fix $\varepsilon \in (0,1)$. For large X let $v^*(X)$ be the first zero of M that is $\ge X$. If $M(\lfloor X \rfloor) \ne 0$, the excursion containing $\lfloor X \rfloor$ starts at some $u < X$ and ends at $v^*(X)$; by Corollary 9.8 with $A = 1+\varepsilon$ (the finitely many earlier excursions lie below X once X is large), $v^*(X) \le u^{1+\varepsilon} < X^{1+\varepsilon}$. Every $n \le X$ with $M(n) \ne 0$ lies in a completed excursion ending at most $v^*(X)$, so, with $\mathfrak{b}_\nu \le H_\nu \ell_\nu$ and $H_\nu \le \max_{n \le v^*} |M(n)| =: m^*$,

$$B_1(X) \le \sum_{v_\nu \le v^*} H_\nu \ell_\nu \le (m^*)^{1/2} \sum_{v_\nu \le v^*} \sqrt{H_\nu}\, \ell_\nu = 2(m^*)^{1/2} \mathcal{A}_{\mathrm{abs}}(v^*),$$

the last equality by Lemma 4.1. By Corollary 8.11 and (8.12), $X^{1+\Theta-\varepsilon} \le B_1(X)$ and $m^* \le (v^*)^{\Theta+\varepsilon} \le X^{(1+\varepsilon)(\Theta+\varepsilon)}$ for large X. Hence $\mathcal{A}_{\mathrm{abs}}(X^{1+\varepsilon}) \ge \mathcal{A}_{\mathrm{abs}}(v^*) \ge \frac{1}{2} X^{1+\Theta/2-3\varepsilon}$. Writing $y = X^{1+\varepsilon}$ and using that $\mathcal{A}_{\mathrm{abs}}$ is nondecreasing, $\mathcal{A}_{\mathrm{abs}}(y) \ge y^{(1+\Theta/2-3\varepsilon)/(1+\varepsilon)-o(1)}$ for all large y. Let $\varepsilon \downarrow 0$. □

Remark 8.14 (Inputs). The lower bound uses the arithmetic localization of returns (Corollary 9.8); for an arbitrary family of disjoint intervals the terminal block of $B_1(X)$ is not controlled by the area up to X. For the lim sup form of the lower bound the classical $B_1(X) = \Omega(X^{1+\Theta-\varepsilon})$ (Mellin transform and Proposition 8.1) suffices instead of [49].

8.2. **Excursion blocks of $1/\zeta$.**

Proposition 8.15 (Excursion blocks of $1/\zeta$)**.** [N-el] *For a complete excursion $I = (x_0, x_1]$ put $D_I(s) = \sum_{x_0 < n \le x_1} \mu(n) n^{-s}$. Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{8.13}$$

For real $\varsigma > 0$, $D_I(\varsigma)$ has the sign of the excursion and

$$|D_I(\varsigma)| \ge \varsigma \mathfrak{b}_I x_1^{-\varsigma-1} \ge \varsigma H_I^2 x_1^{-\varsigma-1}. \tag{8.14}$$

For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{8.15}$$

zero runs contributing nothing.

Proof. For $a_n = \mu(n)$, $A(n) = M(n)$ and $f(n) = n^{-s}$, finite summation by parts gives

$$\sum_{n=x_0+1}^{x_1} a_n f(n) = A(x_1)f(x_1) - A(x_0)f(x_0) + \sum_{n=x_0+1}^{x_1-1} A(n)\big(f(n) - f(n+1)\big).$$

Since $M(x_0) = M(x_1) = 0$, this is (8.13). For real $\varsigma > 0$, all interior $M(n)$ have the excursion sign, and

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} x^{-\varsigma-1}\, dx \geq \varsigma x_1^{-\varsigma-1}.$$

The bound $\mathfrak{b}_I \geq H_I^2$ is (5.14). If M vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every n with $\mu(n) \neq 0$ lies in a completed excursion. For $\Re s > 1$ the ordinary Dirichlet series is absolutely convergent, so summing the finite disjoint blocks gives (8.15). □

Theorem 8.16 (Abscissa of absolute excursion-block convergence)**.** [C] *Let $D_I(s)$ be the completed-excursion blocks of Proposition 8.15. The series*

$$\sum_I D_I(s)$$

has abscissa of absolute convergence, with respect to the excursion blocks, exactly

$$\boxed{\varsigma_{a,\mathrm{exc}} = \Theta}.$$

More precisely, it converges absolutely and locally uniformly for $\Re s > \Theta$ and there represents $1/\zeta(s)$, while for every real $0 < \varsigma < \Theta$ one has

$$\sum_I |D_I(\varsigma)| = \infty.$$

Consequently,

$$\mathrm{RH} \iff \sum_I D_I(s) \text{ converges absolutely for every } \Re s > \tfrac{1}{2}.$$

This is block absolute convergence; the ordinary termwise Dirichlet series $\sum_n \mu(n)n^{-s}$ has abscissa of absolute convergence 1 and, by $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$, abscissa of (conditional) convergence Θ. Grouping by completed excursions converts this conditional convergence into absolute convergence of the blocks.

Proof. Let $s = \varsigma + i\tau$. Since

$$n^{-s} - (n+1)^{-s} = s \int_n^{n+1} x^{-s-1}\, dx,$$

with the real positive branch of $\log x$, one has

$$|n^{-s} - (n+1)^{-s}| \leq |s| n^{-\varsigma-1}.$$

Hence

$$\sum_I |D_I(s)| \leq |s| \sum_{n\geq 1} |M(n)| n^{-\varsigma-1}.$$

Let $K_0 \Subset \{\Re s > \Theta\}$ be compact and set $\varsigma_0 = \min_{s\in K_0} \Re s > \Theta$. Choose $0 < \epsilon_0 < \varsigma_0 - \Theta$. From $B_1(X) = X^{1+\Theta+o(1)}$ we have $B_1(X) \ll_{\epsilon_0} X^{1+\Theta+\epsilon_0}$ for large X, and partial summation gives

$$\sum_{n\geq 1} |M(n)| n^{-\varsigma_0-1} < \infty.$$

Since $\sup_{s\in K_0} |s| < \infty$, the Weierstrass criterion gives normal convergence of the block series on K_0. Thus it is holomorphic in $\Re s > \Theta$. There are no zeros of ζ in that half-plane by the definition of Θ, and $1/\zeta$ extends holomorphically to $s = 1$ with value 0; hence $1/\zeta$ is holomorphic throughout the connected half-plane. The identity already holds for $\Re s > 1$, so the identity theorem extends it to all $\Re s > \Theta$.

For real $\varsigma > 0$, order the completed excursions by their right endpoints. If V is a zero of M, the fixed sign of each excursion and finite Abel summation give

$$\sum_{I:\, x_1(I)\le V} |D_I(\varsigma)| = \sum_{n<V} |M(n)|\big(n^{-\varsigma} - (n+1)^{-\varsigma}\big).$$

Since M has arbitrarily large zeros (Definition 3.1, using $M(x) = \Omega_\pm(\sqrt{x})$), letting $V \to \infty$ through zeros and using monotone convergence gives

$$\sum_{I} |D_I(\varsigma)| = \sum_{n\ge 1} |M(n)|\big(n^{-\varsigma} - (n+1)^{-\varsigma}\big), \tag{8.16}$$

even when both sides are infinite. If the right side were finite for some $0 < \varsigma < \Theta$, then for $n \le X$

$$n^{-\varsigma} - (n+1)^{-\varsigma} = \varsigma \int_n^{n+1} t^{-\varsigma-1}\, dt \ge \varsigma (X+1)^{-\varsigma-1},$$

so

$$B_1(X) \le \frac{(X+1)^{\varsigma+1}}{\varsigma} \sum_{n\ge 1} |M(n)|\big(n^{-\varsigma} - (n+1)^{-\varsigma}\big) = O(X^{1+\varsigma}),$$

contradicting $B_1(X) = \Omega(X^{1+\Theta-\varepsilon})$ for every $\varepsilon > 0$. Therefore the block abscissa is exactly Θ. □

Remark 8.17 (Status of the result). The analytic input is classical: $\sum_n |M(n)| n^{-\varsigma-1} < \infty$ exactly for $\varsigma > \Theta$ (upper bound $M(x) \ll x^{\Theta+\varepsilon}$; lower bound by the Mellin argument of Proposition 8.1). The new point is the excursion grouping: the fixed sign of each completed excursion turns Abel summation into the positive identity (8.16). We did not find this block formulation in the literature (absence of a match is weak evidence).

9. Lacunary excursion families and horizon-localized criteria

9.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 12). For a family $\mathcal{F}$ of excursions and $X \ge 2$ put

$$E_{\mathcal{F}}(X) := \#\{\nu \in \mathcal{F} : x_{0,\nu} \le X\}.$$

For a function $\mathfrak{h} : [0,\infty) \to [0,\infty]$, the family of excursions *crossing the horizon* h is

$$\mathcal{F}_{\mathfrak{h}} := \{\nu : x_{1,\nu} > \mathfrak{h}(x_{0,\nu})\}.$$

For $A > 1$ the excursion I_ν is *A-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$, i.e. $\nu \in \mathcal{F}_{x^A}$, and *$A$-ordinary* otherwise; for $A = 2$ we say *square-exceptional* and *square-ordinary*. For $\lambda > 1$ it is *λ-long* if $x_{1,\nu} > \lambda x_{0,\nu}$. We write $E_A = E_{\mathcal{F}_{x^A}}$, $E_\square = E_2$ and $E_\lambda = E_{\mathcal{F}_{\lambda x}}$; the context separates E_A (A an exponent) from E_λ (λ a factor).

Lemma 9.1 (Power lacunarity)**.** [N-el] *For every $A > 1$ and $X \ge 2$,*

$$E_A(X) \le 2 + \frac{1}{\log A} \log \frac{\log X}{\log 2}. \tag{9.1}$$

In particular

$$E_\square(X) \le 2 + \log_2 \frac{\log X}{\log 2}, \qquad E_\square(X) \le 4 \log\log X \quad (X \ge e^e). \tag{9.2}$$

Proof. No excursion starts at 1, since $M(1) = 1$. List the A-exceptional starts ≥ 2 as $2 \leq a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \geq x_{1,j} > a_j^A$ and $\log a_j \geq A^{j-1} \log 2$. If $a_j \leq X$ then $j \leq 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0, 2]$. For the second bound in (9.2), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \leq 1.443 \log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

Lemma 9.2 (Fractional square horizon)**.** [N-el] *For $0 < c \leq 1$ and $X \geq 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > cx_{0,\nu}^2\} \leq \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{9.3}$$

Consequently, the range $cx_0^2 < t \leq x_0^2$ of the excursions starting at most X is visited by at most $O_c(\log\log X)$ excursions.

Proof. At most $\lceil 2/c \rceil \leq 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \geq 2$. Then $a_{j+1} \geq x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2,$$

and $\log b_j \geq 2^{j-1} \log 2$. Since $b_j \leq cX \leq X$, the number of such j is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \geq t > cx_0^2$. □

Lemma 9.3 (Superlinear horizons)**.** [N-el] *Let $\psi : [1, \infty) \to (0, \infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X).$$

Proof. Given $K > 0$ choose x_K with $\psi(x) \geq e^K$ for $x \geq x_K$. At most $x_K + 1$ members start below x_K. For consecutive members $a_j < a_{j+1}$ starting at or above x_K, $a_{j+1} \geq x_{1,j} > a_j\psi(a_j) \geq e^K a_j$, so $\log a_j$ increases by more than K at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \leq 1/K$ for every K. □

Lemma 9.4 (Linear lacunarity)**.** [N-el] *For every $\lambda > 1$ and $X \geq 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \leq 2 + \frac{\log(X/2)}{\log \lambda}. \tag{9.4}$$

Proof. For starts $a_j \geq 2$, $a_{j+1} \geq x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \geq 2$; add 1 for $(0, 2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

9.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, its translation from $M_0 = M + 2$ to M, and the consequences for excursions.

Theorem 9.5 (Kaczorowski–Pintz [40])**.** [L] *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$, the function M_a of* [40] *at $a = 0$. Let $V(f, Y)$ be defined as in* [40]*, through the largest number of points of $[1, Y]$ at which f is nonzero with alternating signs; the number of sign changes of f on $[1, Y]$ is at least $V(f, Y) - 1$, which does not affect the asymptotic statements below. Then:*

(i) [40, Corollary 4, p. 178, (2.13)–(2.15)]

$$\liminf_{Y\to\infty} \frac{V(M_0, Y)}{\log Y} \geq \frac{\gamma_1}{\pi}, \tag{9.5}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every fixed $\eta > 0$: there exist ordered points $t_1 < \cdots < t_N \leq Y$, with $N \geq (\gamma_1/\pi - o(1)) \log Y$, whose signs alternate and which satisfy $|M_0(t_j)| \geq t_j^{\Theta-\eta}$;

(ii) *The local statement comes from Theorem 1, (2.2)-(2.3), and Corollary 1 of* [40]*; in particular, for every $\delta > 0$ and $\eta > 0$ there is $Y_0(\delta, \eta)$ such that for $Y > Y_0(\delta, \eta)$ the interval $[Y^{1-\delta}, Y]$ contains two points t_+, t_- with $M_0(t_+) \geq t_+^{\Theta-\eta}$ and $M_0(t_-) \leq -t_-^{\Theta-\eta}$. (In* [40] *both losses are governed by a single parameter ε; the form above follows by taking $\varepsilon = \min\{\delta, \eta\}$. Indeed $Y^{1-\varepsilon} \geq Y^{1-\delta}$ and $x^{\Theta-\varepsilon} \geq x^{\Theta-\eta}$, so the original interval and magnitude bound are contained in the formulation above.)*

Lemma 9.6 (From M_0 to M)**.** [N-el] *If $|M_0(t)| > 2$, then $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \geq |M_0(t)| - 2$. In particular, for every fixed $0 < \eta < \Theta$, at points with $|M_0(t)| \geq t^{\Theta-\eta}$ and t large, $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$.*

Proof. $M_0 > 2$ gives $M > 0$; $M_0 < -2$ gives $M < -4$. The second assertion uses $t^{\Theta-\eta} \to \infty$ for $\eta < \Theta$. □

Corollary 9.7 (Distinct large excursions)**.** [N-el] *If $\Theta > 1/2$ and $0 < \eta < \Theta - 1/2$, then*

$$N_\eta(Y) \geq \Big(\frac{\gamma_1}{\pi} - o(1)\Big) \log Y, \tag{9.6}$$

where $N_\eta(Y)$ counts distinct excursions starting at most Y that contain a point $t \leq Y$ with $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$.

Proof. Take the alternating points of Theorem 9.5(i) in $[1, Y]$, discarding the boundedly many below the threshold of Lemma 9.6. By that lemma, consecutive points $t < t'$ have $M(t)$ and $M(t')$ of opposite signs. Since M is integer valued with steps $0, \pm 1$, it vanishes at some integer in (t, t'), so t and t' lie in different excursions. Each point with $M(t) \neq 0$ belongs to exactly one excursion, and excursions are ordered, so the excursions containing the ordered points are pairwise distinct. Their starts are below the points, hence at most Y, and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$ at each point. □

Corollary 9.8 (Multiplicative localization of Mertens returns)**.** [N] *For every fixed $A > 1$ there exists U_A such that every excursion starting at $u \geq U_A$ ends at $v \leq u^A$. Equivalently, M has only finitely many A-exceptional excursions; in particular only finitely many square-exceptional ones.*

Proof. Put $\delta = 1 - 1/A$ and $Y = u^A$, so that $[Y^{1-\delta}, Y] = [u, u^A]$. Fix once and for all $\eta = \Theta/2$. After this choice the resulting threshold depends only on A; denote it by U_A. For $u \geq U_A$, Theorem 9.5(ii) and Lemma 9.6 give $t_+, t_- \in [u, u^A]$ with $M(t_+) > 0 > M(t_-)$. Neither point lies in $[u, u+1)$, because $M(u) = 0$; so $u < \lfloor t_\pm \rfloor$. Since M changes by $0, \pm 1$ at each integer, it vanishes at some integer n with $\min\{\lfloor t_+ \rfloor, \lfloor t_- \rfloor\} < n < \max\{\lfloor t_+ \rfloor, \lfloor t_- \rfloor\}$, and $u < n < u^A$. If $u = x_{0,\nu}$, the first return satisfies $x_{1,\nu} \leq n < u^A$. □

Remark 9.9. A bare sign change of $M_0 = M + 2$ would not force a zero of M. The decisive point is that the local sign change is combined with oscillation of size $t^{\Theta-\eta} \to \infty$, so that Lemma 9.6 applies to both witnesses.

Remark 9.10 (Geometric counts and arithmetic localization). For the actual Mertens excursions Corollary 9.8 is stronger than Lemma 9.1: each A-exceptional family is finite. Consequently, for M itself the power horizons x_0^A and the fractional-square horizons cx_0^2 are inactive for all late excursions, and the corresponding RH-equivalences (Corollary 9.12, Corollary 9.13(i)) follow already from Corollary 9.8; the horizons with genuine content for M are $x_0\psi(x_0)$ and λx_0. The geometric lemmas remain separate results because they are elementary, independent of ζ, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 12 use: the former needs only a count below $(\gamma_1/\pi)\log X$, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

9.3. **Horizon localization.**

Theorem 9.11 (Horizon-localized RH criterion)**.** [N] *Let $\mathfrak{h} : [0,\infty) \to [0,\infty]$ be a function, let $\mathcal{F}$ be a family of excursions, and suppose that for some $\kappa < \gamma_1/\pi$*

$$E_{\mathcal{F}\cup\mathcal{F}_{\mathfrak{h}}}(X) \le (\kappa + o(1))\log X \qquad (X\to\infty). \tag{9.7}$$

Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \le C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t\in I_\nu,\ \nu\notin\mathcal{F},\ t\le \mathfrak{h}(x_{0,\nu}). \tag{9.8}$$

No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $\mathfrak{h}(x_0)$.

Proof. RH implies (9.8) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta,\varepsilon > 0$ with $1/2+\varepsilon < \Theta - \eta$. By (9.6) there are at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting at most X, each containing a witness t with $|M(t)| \ge \frac{1}{2}t^{\Theta-\eta}$. Consider such an excursion ν with witness t. If $\nu\notin\mathcal{F}$ and $t\le\mathfrak{h}(x_{0,\nu})$, the witness lies in the range of (9.8). Otherwise $\nu\in\mathcal{F}$, or $t > \mathfrak{h}(x_{0,\nu})$, in which case $x_{1,\nu} > t > \mathfrak{h}(x_{0,\nu})$ (as $M(t)\neq 0$) and $\nu\in\mathcal{F}_{\mathfrak{h}}$. By (9.7) at least

$$\Big(\frac{\gamma_1}{\pi} - \kappa - o(1)\Big)\log X \longrightarrow \infty$$

witnesses lie in the range of (9.8). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon}\to\infty$, contradicting (9.8). □

The case $\mathfrak{h}(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

Corollary 9.12 (Square horizon)**.** [N] *For every fixed $A > 1$, RH is equivalent to* (9.8) *restricted to A-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2,} \tag{9.9}$$

A separate application of Theorem 9.11, with no excursions discarded and horizon $\mathfrak{h}(x) = x^2$, gives the second RH-equivalent statement

$$|M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad (x_{0,\nu} < t \le \min\{x_{1,\nu}, x_{0,\nu}^2\})$$

over all excursions. The two restricted statements are therefore two distinct consequences of the horizon theorem, not an elementary equivalence between the restrictions themselves.

Proof. Take $\mathcal{F} = \mathcal{F}_{x^A}$, $\mathfrak{h} = \infty$, respectively $\mathcal{F} = \emptyset$, $\mathfrak{h}(x) = x^2$, in Theorem 9.11; Lemma 9.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

Corollary 9.13 (Descending horizons)**.** [N] *RH is equivalent to Littlewood's bound $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ on the ranges*

$$x_{0,\nu} < t \le \min\{x_{1,\nu}, \mathfrak{h}(x_{0,\nu})\}$$

over all excursions, for each of the following horizons:

(i) $\mathfrak{h}(x) = cx^2$, *for any fixed $c > 0$;*
(ii) $\mathfrak{h}(x) = x\psi(x)$, *for any ψ with $\psi(x)\to\infty$;*
(iii) $\mathfrak{h}(x) = \lambda x$, *for any fixed $\lambda > \lambda_* := e^{\pi/\gamma_1} = 1.2489\ldots$*

The same holds with "truncate at $\mathfrak{h}(x_0)$" replaced by "discard the excursions with $x_1 > \mathfrak{h}(x_0)$". In particular it is enough to control the excursions with $x_1 \le \frac{5}{4}x_0$.

Proof. Apply Theorem 9.11 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_{\mathfrak{h}}$, $\mathfrak{h} = \infty$). For (i), $c \geq 1$ reduces to Lemma 9.1 and $0 < c < 1$ to Lemma 9.2; $\kappa = 0$. For (ii), Lemma 9.3 gives $\kappa = 0$. For (iii), Lemma 9.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < \gamma_1/\pi$ exactly when $\lambda > \lambda_*$. Finally $5/4 > \lambda_*$. □

At the level of the restricted statements, (i) and (ii) imply (iii) eventually, because their horizons are at least λx_0 for every fixed admissible $\lambda > \lambda_*$ once x_0 is large. The converse does not follow from inclusion of ranges; rather, each of (i)–(iii) is separately RH-equivalent by Theorem 9.11. The finitely many small excursions affect only the constant C_ε. The horizon has been lowered from x_0^2 to λx_0: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

Corollary 9.14 (Two-sided localization of large values)**.** [N] *Suppose $\Theta > 1/2$, and fix $0 < \eta < \Theta - \frac{1}{2}$ and $\lambda > \lambda_*$ before letting $X \to \infty$. Among the excursions starting at $x_0 \leq X$, at least*

$$\Big(\frac{\gamma_1}{\pi} - \frac{1}{\log\lambda} - o(1)\Big)\log X$$

contain a point t_* with $|M(t_*)| \geq \frac{1}{2}t_*^{\Theta-\eta}$ and

$$x_0 + \tfrac{1}{2}x_0^{\Theta-\eta} \leq t_* \leq \lambda x_0.$$

Proof. Of the excursions of Corollary 9.7, those whose witness exceeds λx_0 are λ-long, hence at most $\log X/\log\lambda + 2$ by Lemma 9.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \leq t_* - x_0$, so $t_* - x_0 \geq \frac{1}{2}t_*^{\Theta-\eta} \geq \frac{1}{2}x_0^{\Theta-\eta}$. □

Remark 9.15 (Dyadic erosion and the uniform transition). Fix an integer $j \geq 0$ before letting $X \to \infty$. Lemma 9.2 with $c = 2^{-j}$ gives

$$\#\left\{\nu : x_{0,\nu} \leq X,\ x_{1,\nu} > \frac{x_{0,\nu}^2}{2^j}\right\} = O_{2^j}(\log\log X),$$

so Theorem 9.11 successively permits the horizons

$$x_0^2, \quad \frac{x_0^2}{2}, \quad \frac{x_0^2}{4}, \quad \frac{x_0^2}{8}, \dots.$$

The constant in the fixed-c estimate depends on c; hence one may not set $j = j(x_0) \sim \log_2 x_0$ by that lemma alone. If instead $2^{j(x)} = x/\psi(x)$ with $\psi(x) \to \infty$, then the corresponding horizon is $x\psi(x)$ and Lemma 9.3 supplies the uniform $o(\log X)$ crossing count. Thus the rigorous erosion chain is

$$x_0^2 \to \frac{x_0^2}{2} \to \frac{x_0^2}{4} \to \cdots \to x_0\psi(x_0) \to \lambda x_0, \qquad \lambda > \lambda_*.$$

9.4. **Fixed-mask Catalan compression inside fractional square horizons.** The fractional-square localization has a useful interaction with the exact combinatorics of a completed excursion. The relevant Catalan time is not physical time: zero Möbius increments must first be removed. This support-time distinction is essential below.

Definition 9.16 (Support-time path and fixed-mask Catalan law)**.** Let $I = (x_0, x_1]$ be a complete excursion, and list the nonzero increments in I as

$$x_0 < a_1 < a_2 < \cdots < a_L \leq x_1, \qquad \mu(a_j) \neq 0.$$

Put

$$r_I(t) := \#\{j : a_j \leq t\}, \qquad x_0 < t \leq x_1.$$

For the excursion I_ν, let $\sigma_\nu \in \{\pm 1\}$ be its orientation and define

$$S_0 = 0, \qquad S_r := \sigma_\nu \sum_{j=1}^{r} \mu(a_j) \qquad (1 \leq r \leq L).$$

Then $L = 2n$ is even and

$$S_0 = S_L = 0, \qquad S_r > 0 \quad (1 \le r < L), \qquad S_r - S_{r-1} \in \{-1, +1\}.$$

Thus $(S_r)_{0 \le r \le L}$ is a primitive Dyck path. The *fixed-mask primitive Catalan law* keeps the physical support $\{a_1, \dots, a_L\}$ fixed and assigns to those support positions a uniformly chosen primitive Dyck sign sequence of length L. We denote its probability by $\mathbb{P}_{\mathrm{pCat},I}$.

For the observed excursion one has, exactly,

$$|M(t)| = S_{r_I(t)}, \qquad r_I(t) \le t - x_0 < t. \tag{9.10}$$

The first equality uses the orientation of the excursion and the fact that M is constant across the zero increments.

We first record the global Catalan estimate, since it is useful in its own right and clarifies why a different estimate is needed at the square horizon.

Lemma 9.17 (Catalan maximum tail)**.** [L] *Let $L = 2n$ and let $H_{\max}$ be the maximum of a uniformly chosen Dyck path of length $2n$. For $1 \le h \le n$,*

$$\mathbb{P}_{\mathrm{Cat},L}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le (n+1)\exp\left(-\frac{h^2}{n+h}\right) \le \left(\frac{L}{2}+1\right)e^{-h^2/L}. \tag{9.11}$$

For a uniformly chosen primitive Dyck path of length $L = 2n$,

$$\mathbb{P}_{\mathrm{pCat},L}\{H_{\max} \ge h\} \le \frac{L}{2}\exp\left(-\frac{(h-1)^2}{L-2}\right), \qquad 2 \le h \le n. \tag{9.12}$$

Proof. There are $\mathrm{Cat}_n = (n+1)^{-1}\binom{2n}{n}$ Dyck paths of length $2n$ and $\binom{2n}{n}$ unrestricted bridges. Reflecting a bridge after its first visit to level h gives a bijection between bridges that reach h and unrestricted walks ending at $2h$, hence their number is $\binom{2n}{n+h}$. Therefore

$$\mathbb{P}_{\mathrm{Cat},2n}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}}.$$

Moreover

$$\frac{\binom{2n}{n+h}}{\binom{2n}{n}} = \prod_{j=1}^{h}\frac{n-j+1}{n+j},$$

and $\log(1-x) \le -x$ gives

$$\log\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le -\sum_{j=1}^{h}\frac{2j-1}{n+j} \le -\frac{h^2}{n+h}.$$

Since $n+h \le 2n = L$, (9.11) follows. A primitive path of length $2n$ has the form $+1$, followed by a Dyck path of length $2n-2$, followed by -1; its maximum is one plus the maximum of the interior Dyck path. Applying the preceding bound with $n-1$ and $h-1$ proves (9.12). □

Remark 9.18 (Why the maximum tail is not the square-horizon estimate). The exponent in (9.12) is of order H^2/L. For a general family of disjoint intervals retained by the square horizon one may have $L \asymp x_0^2$ (for M itself such excursions are finitely many by Corollary 9.8), while the relevant witness lies much earlier, at physical time $t \asymp x_0$. Then a violation of size $t^{1/2+\varepsilon}$ need not make H^2/L large. Thus the global maximum tail, although correct, is not the right Catalan quantity for combining with the fractional-square horizon. The correct quantity is the pointwise height at the support time $r_I(t)$.

Lemma 9.19 (Pointwise primitive-Catalan tail)**.** [N-el] *Let $L = 2n \geq 4$ and let (S_r) be a uniformly chosen primitive Dyck path of length L. For $1 \leq r \leq L-1$ and real $h \geq 1$,*

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \geq h\} \leq 8n^{3/2} \exp\left(-\frac{h^2}{2r}\right) \leq 3L^{3/2} \exp\left(-\frac{h^2}{2r}\right). \tag{9.13}$$

By time reversal the same estimate holds with r replaced by $L-r$; hence

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \geq h\} \leq 3L^{3/2} \exp\left[-\frac{h^2}{2\min\{r, L-r\}}\right]. \tag{9.14}$$

If $h > \min\{r, L-r\}$ the probability is 0, so the bound is only relevant for $h \leq \min\{r, L-r\}$.

Proof. Ignore both positivity and the endpoint constraint. Among all 2^L sign words, the proportion whose first r signs have sum at least h is at most $\exp(-h^2/(2r))$ by the elementary Chernoff bound for a Rademacher sum. Therefore the number of primitive Dyck paths satisfying $S_r \geq h$ is at most

$$2^L \exp\left(-\frac{h^2}{2r}\right).$$

The total number of primitive Dyck paths of length $2n$ is Cat_{n-1}. The standard central-binomial lower bound $\binom{2m}{m} \geq 4^m/(2\sqrt{m})$ gives

$$\mathrm{Cat}_{n-1} = \frac{1}{n}\binom{2n-2}{n-1} \geq \frac{4^{n-1}}{2n\sqrt{n-1}}.$$

Since $2^L = 4^n$, division gives the first inequality in (9.13); the second follows from $8n^{3/2} < 3(2n)^{3/2}$. Reversing the path and changing the signs of the reversed increments preserves the primitive Dyck class and carries height S_r to height S_{L-r}, which proves (9.14). □

For an observed excursion I define its pointwise Catalan upper-tail value by

$$p_{\mathrm{Cat}}(I,t) := \mathbb{P}_{\mathrm{pCat},I}\left\{S_{r_I(t)} \geq |M(t)|\right\}. \tag{9.15}$$

This is a tail probability under the fixed-mask Catalan reference law; it is not a probability assigned to the arithmetic Möbius path itself.

Corollary 9.20 (Fractional-square Catalan rarity)**.** [N-el] *Fix $c > 0$ and $\varepsilon > 0$. Let $I = (x_0, x_1]$ be a complete excursion with $x_1 \leq cx_0^2$, and keep its zero mask fixed. For all sufficiently large x_0,*

$$\mathbb{P}_{\mathrm{pCat},I}\left\{\exists\, t \in I \cap \mathbb{Z} :\ S_{r_I(t)} \geq \lceil t^{1/2+\varepsilon} \rceil\right\} \leq 3c^{5/2} x_0^5 \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right). \tag{9.16}$$

In particular the right-hand side is smaller than x_0^{-A} for every fixed $A > 0$ once x_0 is large enough.

Proof. At a support position $t = a_r$ one has $r \leq t - x_0 < t$. Hence Lemma 9.19 with $h = \lceil t^{1/2+\varepsilon} \rceil$ gives

$$\mathbb{P}_{\mathrm{pCat},I}\{S_r \geq t^{1/2+\varepsilon}\} \leq 3L^{3/2} \exp\left(-\frac{1}{2}t^{2\varepsilon}\right) \leq 3L^{3/2} \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right).$$

There are at most L support positions. Since $L \leq x_1 - x_0 \leq x_1 \leq cx_0^2$, one has $3L \cdot L^{3/2} = 3L^{5/2} \leq 3(cx_0^2)^{5/2} = 3c^{5/2}x_0^5$, and the union bound gives (9.16). Between support positions the path is constant, so no additional physical times have to be counted. □

The preceding statement is purely combinatorial: it measures how far a large value lies in the tail of a reference law, not how the Möbius constraints select a path. Combined with the localization theorem it gives the following quantitative form.

Corollary 9.21 (Catalan cost of a Kaczorowski–Pintz witness under $\Theta > 1/2$)**.** [N] *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, put $\delta = \Theta - \eta - \frac{1}{2}$, and fix $\lambda > \lambda_*$. Then there are infinitely many excursions $I = (x_0, x_1]$ with $x_1 \le \lambda x_0$ containing a point t with*

$$-\log p_{\mathrm{Cat}}(I,t) \;\ge\; \frac{1}{8}x_0^{2\delta} - O(\log x_0). \tag{9.17}$$

The same holds with $x_1 \le \lambda x_0$ replaced by $x_1 \le c x_0^2$ for any fixed $c > 0$.

Proof. By Corollary 9.7 and Lemma 9.4, at least $(\gamma_1/\pi - 1/\log\lambda - o(1))\log X$ excursions starting at most X have $x_1 \le \lambda x_0$ and contain t with $h = |M(t)| \ge \frac{1}{2}t^{1/2+\delta}$; for cx_0^2 use Lemma 9.2 instead. With $r = r_I(t) < t$ and $L \le x_1 - x_0$, Lemma 9.19 gives $p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp(-h^2/(2r)) \le 3L^{3/2}\exp(-t^{2\delta}/8)$, and $t > x_0$, $L \le \lambda x_0$ (respectively $L \le cx_0^2$). □

Remark 9.22 (Selector bounds are one-sided). Lemma 9.19 gives only the upper-tail estimate

$$p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp\!\left(-\frac{|M(t)|^2}{2r_I(t)}\right),$$

and hence the one-sided implication

$$-\log p_{\mathrm{Cat}}(I,t) \ge \frac{|M(t)|^2}{2r_I(t)} - \log(3L^{3/2}).$$

No reverse inequality, and therefore no equivalence "up to polynomial factors", is proved here. Consequently a selector bound $-\log p_{\mathrm{Cat}}(I,t) \le \Phi(x_0)$ with $\Phi(x) = x^{o(1)}$ on excursions with $x_1 \le \lambda x_0$ implies $|M(t)| \le t^{1/2}x_0^{o(1)}$ there, so RH follows from Corollary 9.13; this is a sufficient condition, not an independent equivalent formulation. A polynomial lower bound $p_{\mathrm{Cat}} \ge x_0^{-A}$ would similarly give $|M(t)| \ll \sqrt{t\log t}$ on the retained ranges. To turn the Catalan weight into a two-sided measure of deviation one would need a matching lower-tail estimate of the form $-\log p_{\mathrm{Cat}}(I,t) \ll |M(t)|^2/r_I(t) + \log L$, which is not proved here.

Corollary 9.23 (Catalan cost along a Littlewood-violating sequence)**.** [N-el] *Let $t_j \to \infty$, $A_j \to \infty$ and $\varepsilon_0 > 0$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$, and let $I_j = (x_{0,j}, x_{1,j}]$ be the excursion containing t_j. If $x_{1,j} \le x_{0,j}^B$ for a fixed B, then*

$$-\log p_{\mathrm{Cat}}(I_j,t_j) \;\ge\; \frac{A_j^2}{2}\,t_j^{2\varepsilon_0} - O_B(\log x_{0,j}) \;\ge\; \frac{A_j^2}{2}\,x_{0,j}^{2\varepsilon_0} - O_B(\log x_{0,j}).$$

Proof. With $r = r_{I_j}(t_j) < t_j$ and $h = |M(t_j)|$, $h^2/(2r) \ge A_j^2 t_j^{1+2\varepsilon_0}/(2t_j)$; apply Lemma 9.19 with $L \le x_{1,j} \le x_{0,j}^B$. □

Selection energies: complete excursions and open prefixes. The support-time clock of Definition 9.16 is the squarefree-compressed clock

$$\mathcal{S}(x) = \#\{n \le x : \mu(n) \ne 0\}, \qquad L^{\#}(I) = \mathcal{S}(x_1) - \mathcal{S}(x_0) \le x_1 - x_0,$$

and $r_I(t) = \mathcal{S}(t) - \mathcal{S}(x_0)$. Two tail quantities are natural: one for the height of a complete excursion, and one for an open prefix, which does not require knowledge of the future return.

Definition 9.24 (Selection energies)**.** For a complete excursion I with height $H(I)$ and compressed length $L^{\#} = L^{\#}(I)$, put

$$p_{\mathrm{Cat}}^{\max}(I) := \mathbb{P}_{\mathrm{pCat},L^{\#}}\{H_{\max} \ge H(I)\}, \qquad \mathcal{I}_{\mathrm{Cat}}(I) := -\log p_{\mathrm{Cat}}^{\max}(I).$$

For $t \ge 1$ with $M(t) \ne 0$ let $a(t)$ be the last zero of M before t and $r = \mathcal{S}(t) - \mathcal{S}(a(t))$. After orientation, the compressed segment from $a(t)$ to t is a positive meander of length r: a ± 1 path

$(S_k)_{k\le r}$ with $S_0=0$ and $S_k>0$ for $1\le k\le r$. Let $\mathcal{M}_r$ be the set of such paths. Since $S_r\equiv r \pmod 2$, an endpoint event with incompatible parity has probability zero. Define

$$p_{\mathrm{open}}(t) := \mathbb{P}_{\mathcal{M}_r}\{S_r \ge |M(t)|\},$$

the uniform measure on $\mathcal{M}_r$.

Lemma 9.25 (Height–entropy inequalities)**.** [N-el]

(i) *If $L^{\#}=2$, the compressed path is $0,1,0$ and $H(I)=1$. For every complete excursion with $L^{\#}\ge 4$,*

$$\mathcal{I}_{\mathrm{Cat}}(I) \;\ge\; \frac{(H(I)-1)^2}{L^{\#}-2} - \log\frac{L^{\#}}{2}.$$

(ii) $\#\mathcal{M}_r = \binom{r-1}{\lfloor (r-1)/2\rfloor} \ge 2^r/(4\sqrt{r})$*, and for $h\ge 1$*

$$\mathbb{P}_{\mathcal{M}_r}\{S_r\ge h\} \le 4\sqrt{r}\,e^{-h^2/(2r)}, \qquad -\log p_{\mathrm{open}}(t) \;\ge\; \frac{M(t)^2}{2r} - \log\bigl(4\sqrt{r}\bigr).$$

Proof. (i) is (9.12). (ii) The count of positive meanders is classical (ballot theorem; the number ending at s is $\frac{s}{r}\binom{r}{(r+s)/2}$). From $\binom{2m}{m}\ge 4^m/(2\sqrt{m})$ and $\binom{2m+1}{m}=\frac12\binom{2m+2}{m+1}$ one gets $\#\mathcal{M}_r\ge 2^r/(4\sqrt{r})$. Among all 2^r words the proportion with $S_r\ge h$ is at most $e^{-h^2/(2r)}$ by the Chernoff bound; divide. □

Proposition 9.26 (Selector criteria)**.** [N]

(i) Complete excursions. *Fix $\lambda>\lambda_*$. If there is A such that $p^{\max}_{\mathrm{Cat}}(I)\ge x_0^{-A}$ for every sufficiently large complete excursion with $x_1\le \lambda x_0$, then $H(I)\ll_{A,\lambda}\sqrt{x_0\log x_0}$ on that family, and RH holds.*

(ii) Open prefixes. *If there is A such that $p_{\mathrm{open}}(t)\ge t^{-A}$ for all large t with $M(t)\ne 0$, then $|M(t)|\ll_A \sqrt{t\log t}$, and RH holds.*

Proof. (i) If $L^{\#}=2$ then $H(I)=1$. Otherwise $L^{\#}\ge 4$ and, by Lemma 9.25(i), $(H-1)^2\le (L^{\#}-2)(A\log x_0+\log L^{\#})$, and $L^{\#}\le(\lambda-1)x_0$. Littlewood's bound therefore holds on every excursion with $x_1\le\lambda x_0$, and Corollary 9.13(iii), in the form that discards the λ-long excursions, gives RH. (ii) By Lemma 9.25(ii), $M(t)^2\le 2r(A\log t+\log(4\sqrt{r}))$ with $r\le t$. □

Remark 9.27 (The threshold λ_*). The function $\cos(\gamma_1\log x)$ changes sign γ_1/π times per unit of $\log x$, i.e. once in every multiplicative interval of ratio $e^{\pi/\gamma_1}=\lambda_*$; the constant in (9.5) is this density. On the geometric side, a chain of λ-long intervals realizes the bound of Lemma 9.4. The method of Theorem 9.11 compares these two counts, and λ_* is the exact point where they balance: for $\lambda\le\lambda_*$ the λ-long family can, as far as counting alone is concerned, absorb every Kaczorowski–Pintz excursion. Going below λ_* requires information on where inside an excursion the large values occur, not only how many excursions carry them.

Proposition 9.28 (Logarithmic-length budget)**.** [N-el] *Let $\mathcal{F}$ be a family of excursions and let $u_1<\cdots<u_N$ be the starts of its members with $2\le u_j\le X$, with ends v_j. The initial excursion $(0,2]$ is explicitly excluded, since $\log(v/u)$ is not defined at $u=0$. Then*

$$\sum_{j=1}^{N-1}\log\frac{v_j}{u_j}\le\log\frac{X}{2}.$$

If moreover $v_N\le u_N^A$ for some $A\ge 1$, then $\sum_{j=1}^{N}\log(v_j/u_j)\le A\log X$.

Proof. Disjointness gives $u_{j+1}\ge v_j$, so $\log(v_j/u_j)\le\log u_{j+1}-\log u_j$; the sum telescopes to at most $\log u_N-\log u_1\le\log(X/2)$. The last term is at most $(A-1)\log u_N\le(A-1)\log X$. □

Lemmas 9.1–9.4 are the special cases in which every term is bounded below. The budget form shows what the horizon theorem actually measures.

Corollary 9.29 (Mean logarithmic length of Kaczorowski–Pintz excursions)**.** [N] *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, and let $\mathcal{K}(X)$ be any set of distinct excursions starting at most X, each containing a point t with $|M(t)| \geq \frac{1}{2}t^{\Theta-\eta}$, such that $\#\mathcal{K}(X) \geq (\gamma_1/\pi - o(1))\log X$ (such sets exist by Corollary 9.7). Then*

$$\frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \;\leq\; \frac{\pi}{\gamma_1} + o(1) = \log \lambda_* + o(1).$$

Proof. Discard the initial excursion $(0, 2]$, whose start is 0 and whose contribution to the count is $O(1)$. Fix $A > 1$ before taking $X \to \infty$. By Corollary 9.8, all but finitely many excursions satisfy $v \leq u^A$. These exceptions form a fixed global finite set once A is fixed, so the sum of their individual finite values $\log(v/u)$ is $O_A(1)$. Proposition 9.28 gives a total of at most $A \log X + O_A(1)$. Since $\#\mathcal{K}(X) \geq (\gamma_1/\pi - o(1))\log X$, it follows for every fixed $A > 1$ that

$$\limsup_{X\to\infty} \frac{1}{\#\mathcal{K}(X)} \sum_{\nu \in \mathcal{K}(X)} \log \frac{v_\nu}{u_\nu} \leq \frac{A\pi}{\gamma_1}.$$

Only after this limit is taken do we let $A \downarrow 1$, obtaining π/γ_1. □

Remark 9.30 (What lowering λ_* requires). Corollary 9.13(iii) is the statement that the excursions carrying Kaczorowski–Pintz witnesses cannot all have $\log(v/u) > \log\lambda$ with $\lambda > \lambda_*$; Corollary 9.29 is its averaged form. Consequently, any proof that such excursions have mean logarithmic length at least $\pi/\gamma_1 + \delta$ for some fixed $\delta > 0$ would contradict Corollary 9.29 and prove RH, and a surplus $\delta > \log(\lambda_*/\lambda)$ among the excursions whose witness lies beyond λx_0 would extend Corollary 9.13(iii) to that $\lambda < \lambda_*$. Kaczorowski–Pintz, as stated, gives neither: it supplies at least one return in each band $[Y^{1-\delta}, Y]$ but no upper separation between consecutive witnesses. Nor do short-interval estimates for μ supply it: $|M(x+h) - M(x)| = o(h)$ is compatible with M staying positive throughout $(x, x+h]$, whereas a return requires a decrement of size at least $M(x)$ in a prescribed direction. Energy bounds do not help either, since a flat excursion of height 1 has $\mathfrak{b}_I, Q_I \asymp \ell_I$.

Wide peaks from cancellation in all short intervals. The only arithmetic input of the following lemma is the quantitative short-interval theorem of Matomäki–Teräväinen [44, Theorem 1.1]: for every fixed $\theta > 0.55$ and every fixed $\varepsilon > 0$, uniformly for the relevant intervals,

$$\sum_{x < n \leq x+H} \mu(n) \ll_{\theta,\varepsilon} \frac{H}{(\log x)^{1/3-\varepsilon}} \qquad (H \geq x^\theta).$$

Lemma 9.31 (Mesoscopic Lipschitz bound)**.** [C] *Fix $\Lambda > 1$ and $\theta_{\mathrm{w}} > 0.55$. For large u put $h_u = \lceil (\Lambda u)^{\theta_{\mathrm{w}}} \rceil$ and*

$$\omega_u := \max\Big\{ h_u^{-1},\ \max_{u \leq a \leq \Lambda u} \frac{|M(a+h_u) - M(a)|}{h_u} \Big\}.$$

Then $\omega_u \to 0$, and for all integers $u \leq a < b \leq \Lambda u$,

$$|M(b) - M(a)| \leq \omega_u (b-a) + h_u. \tag{9.18}$$

Proof. Fix $\theta' \in (0.55, \theta_{\mathrm{w}})$. For all sufficiently large u, $a + h_u \leq 2\Lambda u$ uniformly for $u \leq a \leq \Lambda u$. By [44, Theorem 1.1], for every fixed $\varepsilon > 0$ there is a constant C such that, uniformly for $u \leq b \leq 2\Lambda u$,

$$\Big| M\Big(b + \lceil b^{\theta'} \rceil\Big) - M(b) \Big| \leq C \frac{b^{\theta'}}{(\log u)^{1/3-\varepsilon}}$$

for all sufficiently large u (changing C absorbs the harmless rounding).

For a given a, define recursively

$$b_0 = a, \qquad b_{j+1} = b_j + \lceil b_j^{\theta'} \rceil,$$

and stop at the first J for which $b_{J+1} > a + h_u$. Since $b_j \asymp u$ throughout the construction, each full block has length $\asymp u^{\theta'}$, whence

$$J \ll \frac{h_u}{u^{\theta'}} + 1, \qquad \sum_{j<J} b_j^{\theta'} \ll h_u.$$

All starting points b_j lie in $[u, 2\Lambda u]$. Summing the full-block estimates and bounding the final incomplete piece trivially gives, uniformly for $u \le a \le \Lambda u$,

$$\frac{|M(a+h_u) - M(a)|}{h_u} \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}}.$$

Consequently

$$\omega_u \ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda} (\log u)^{-1/3+\varepsilon} + u^{\theta'-\theta_{\mathrm{w}}} + h_u^{-1} = o(1), \tag{9.19}$$

and the same estimate is uniform for $u \in [U, 2U]$ as $U \to \infty$.

For (9.18), write $b - a = qh_u + r$ with $0 \le r < h_u$. Every complete h_u-block starts at a point in $[u, \Lambda u]$ and contributes at most $\omega_u h_u$, while the terminal remainder contributes at most $r < h_u$. Thus $|M(b) - M(a)| \le \omega_u(b-a) + h_u$. □

Proposition 9.32 (Wide peaks)**.** [N] *Let $(u, v]$ be an excursion with $v \le \Lambda u$, let $u < t < v$ with $|M(t)| = H$, and put $G = (H - h_u)_+$ and $m = \lfloor G/\omega_u \rfloor$. Then*

$$t - u \ge \frac{G}{\omega_u}, \qquad v - t \ge \frac{G}{\omega_u},$$

$$\mathfrak{b} \ge H + 2\sum_{j=1}^{m}(G - \omega_u j), \qquad Q \ge H^2 + 2\sum_{j=1}^{m}(G - \omega_u j)^2.$$

Consequently, uniformly for such excursions with $H/h_u \to \infty$,

$$v - u \ge (2 + o(1))\frac{H}{\omega_u}, \qquad \mathfrak{b} \ge (1 + o(1))\frac{H^2}{\omega_u}, \qquad Q \ge \Big(\frac{2}{3} + o(1)\Big)\frac{H^3}{\omega_u},$$

so that $(v-u)/H$, $\mathfrak{b}/H^2$ and Q/H^3 tend to infinity, whereas the unit-step bounds of Lemma 3.3 and Proposition 5.6 give only 2, 1 and $\frac{2}{3}$.

Proof. By (9.18) applied to $(u, t]$ and $(t, v]$, $H \le \omega_u(t-u) + h_u$ and $H \le \omega_u(v-t) + h_u$. Thus, since the two distances are integers,

$$t - u,\ v - t \ge \left\lceil \frac{G}{\omega_u} \right\rceil,$$

and therefore $m = \lfloor G/\omega_u \rfloor \le t - u, v - t$. For $1 \le j \le m$ the points $t \pm j$ lie in $[u, v]$, and (9.18) gives $\sigma_\nu M(t \pm j) \ge G - \omega_u j \ge 0$. Summing t and the pairs $t \pm j$ gives the bounds for $\mathfrak{b}$ and Q. Finally $2\sum_{j \le m}(G - \omega_u j) = G^2/\omega_u + O(G)$ and $2\sum_{j\le m}(G-\omega_u j)^2 = \frac{2}{3}G^3/\omega_u + O(G^2)$. Under $H/h_u \to \infty$ we have $G = H - h_u = (1 + o(1))H$ and $G \to \infty$. Equation (9.19) gives $\omega_u = o(1)$, hence eventually $\omega_u \le 1$ and the relative errors satisfy $\omega_u/G \le 1/G = o(1)$. This proves the stated uniform asymptotics. □

Corollary 9.33 (Packing of high excursions)**.** [N] *Let $Y = Y(U)$ with $Y/U_{\mathrm{w}}^{\theta} \to \infty$. The number of excursions with $U \le u \le 2U$, $v \le \Lambda u$ and height at least Y is $o(U/Y)$, whereas the unit-step bound gives $O(U/Y)$.*

Proof. They are disjoint and contained in $[U, 2\Lambda U]$. By Proposition 9.32, each has length at least

$$(2+o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u}.$$

If $N(U,Y)$ denotes their number, disjointness therefore gives

$$N(U,Y)(2+o(1))\frac{Y}{\max_{U\le u\le 2U}\omega_u}\le (2\Lambda-1)U.$$

By the uniform estimate (9.19),

$$\omega_U^* := \max_{U\le u\le 2U}\omega_u = o(1).$$

Therefore

$$N(U,Y)\ll_\Lambda \omega_U^*\frac{U}{Y}=o(U/Y),$$

which completes the packing argument. □

Remark 9.34 (Scope). Proposition 9.32 is informative only for heights above $u^{0.55+o(1)}$. If $\Theta>0.55$ and $0.55<\theta_{\mathrm{w}}<\Theta-\eta$, it applies to the Kaczorowski–Pintz witness excursions of Corollary 9.14 with $v\le\lambda u$: their peaks are wide in the sense above. It does not lower λ_*: the resulting lower bound for $\log(v/u)$ is of order $H/(\omega_u u)$, which may still tend to 0, whereas Remark 9.30 requires a fixed surplus. The published quantitative estimate of Matomäki–Teräväinen [44, Theorem 1.1] makes the gain explicit. If $0.55<\theta'<\theta_{\mathrm{w}}$, then the blocking argument in Lemma 9.31 gives, for every fixed $\varepsilon>0$,

$$\omega_u\ll_{\varepsilon,\theta',\theta_{\mathrm{w}},\Lambda}(\log u)^{-1/3+\varepsilon}+u^{\theta'-\theta_{\mathrm{w}}}.$$

Consequently, in the regime $H/h_u\to\infty$ of Proposition 9.32,

$$Q\gg_\varepsilon H^3(\log u)^{1/3-\varepsilon}.$$

10. Prefix structure of the localized window

10.1. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 10.31 is no longer needed: on $(u,2u]$ the Möbius function is a *linear* expression in the prefix $\mu\mathbf{1}_{[1,u]}$.

Proposition 10.1 (Linear window)**.** [L] *Let $u\ge 1$. For every integer $u<n\le 2u$,*

$$\mu(n)=-\sum_{\substack{d\mid n\\ d\le u}}\mu(d). \tag{10.1}$$

Consequently, if $M(u)=0$ and $u<t\le 2u$,

$$M(t)=-\sum_{d\le u}\mu(d)\left(\left\lfloor\frac{t}{d}\right\rfloor-\left\lfloor\frac{u}{d}\right\rfloor\right). \tag{10.2}$$

Proof. Since $n\ge 2$, $\sum_{d\mid n}\mu(d)=0$. Every proper divisor of n is at most $n/2\le u$, and every divisor $d\le u$ is proper because $n>u$. Summing (10.1) over $u<n\le t$, the number of multiples of d in $(u,t]$ is $\lfloor t/d\rfloor-\lfloor u/d\rfloor$. □

Corollary 10.2 (Localized prefix formula)**.** [N-el] *Let $M(u)=0$, let $1<\lambda<2$, and let $u<t\le\lambda u$. Then*

$$M(t)=-\sum_{d\le\lambda u/2}\mu(d)\left(\left\lfloor\frac{t}{d}\right\rfloor-\left\lfloor\frac{u}{d}\right\rfloor\right). \tag{10.3}$$

Indeed, if $\lambda u/2<d\le u$, then $d>u/2$ and the first multiple of d strictly larger than u is $2d>\lambda u\ge t$.

Corollary 10.3 (Prefix-dependency cutoff)**.** [N] *Put*

$$\chi_* := \frac{1}{2}e^{\pi/\gamma_1} = 0.62445\ldots.$$

For every fixed χ with $\chi_* < \chi < 1$, RH is equivalent to Littlewood's bound on every excursion starting at u throughout

$$u < t \le \min\{x_1, 2\chi u\},$$

where, throughout that window, $M(t)$ is given by the sum (10.3) using only the prefix $\mu(d)$ with $d \le \chi u$. In particular, the choice $\chi = 5/8$ corresponds to $\lambda = 5/4$ and gives the RH-equivalent window

$$1 \le r = t - u \le \min\{\ell - 1, u/4\}$$

using only $d \le 5u/8$. The return point $r = \ell$, when it belongs to the window, is omitted because $M(u + \ell) = 0$.

The number χ_* is an infimum, not an attained endpoint, because the horizon theorem requires $\lambda > \lambda_*$.

Proof. Take $\lambda = 2\chi > \lambda_*$ in Corollary 9.13(iii), then use Corollary 10.2. □

Corollary 10.4 (Algebraic dependency gap)**.** [N-el] *Let $\chi_* < \chi < 1$. If u is a zero of M, then every increment $\mu(n)$ with*

$$u < n \le 2\chi u$$

is determined by the prefix $\mu(1), \ldots, \mu(\lfloor \chi u \rfloor)$. Consequently the intermediate block

$$(\chi u, u]$$

does not occur as a direct input in the divisor recursion determining the localized future window. As $\chi \downarrow \chi_*$, the omitted block has relative length tending to $1 - \chi_* = 0.3755\ldots$.

Proof. For $u < n \le 2\chi u$, every proper divisor $d \mid n$ satisfies $d \le n/2 \le \chi u$. Hence

$$\mu(n) = -\sum_{\substack{d \mid n \\ d < n}} \mu(d)$$

uses only $d \le \chi u$. □

In convolution form, (10.1) says that $(\mu - \mu_u) * 1 = \mu \mathbf{1}_{(u,\infty)}$ on $n \le 2u$, the order-one analogue of (10.10), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (10.2) follows from $\sum_{d \le t} \mu(d)\lfloor t/d \rfloor = 1$, since $\lfloor t/d \rfloor = 1$ for $u < d \le t \le 2u$.

Proposition 10.5 (One-step recursion)**.** [L] *For real $x \ge 1$,*

$$\sum_{k \le x} M\Big(\frac{x}{k}\Big) = 1. \tag{10.4}$$

Hence, if $M(u) = 0$ and $t > u$,

$$M(t) = -\sum_{2 \le k \le t} \left[M\left(\frac{t}{k}\right) - M\Big(\frac{u}{k}\Big) \right], \tag{10.5}$$

with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; hence on a localized window $u < t \le \lambda u$ all arguments are at most $\lambda u/2$. For $\lambda = 5/4$ this is $5u/8$.

Proof. $\sum_{k \le x} M(x/k) = \sum_{k \le x} \sum_{d \le x/k} \mu(d) = \sum_{m \le x} \sum_{d \mid m} \mu(d) = 1$. Here and below $M(x/k) = M(\lfloor x/k \rfloor)$ by the global real-argument convention. Subtract the identity at u, where $M(u) = 0$, and extend the second sum to $k \le t$ using $M(u/k) = 0$ for $k > u$. □

Proposition 10.6 (Scale cascade in the localized excursion window)**.** [N-el] *Let $(u,v]$ be an excursion, let $\lambda_* < \lambda < 2$, and let $u < t \le \min\{v, \lambda u\}$. For $k \ge 2$ put*

$$\Delta_k(u,t) := M(t/k) - M(u/k),$$

and for an integer $K \ge 2$ put

$$\mathcal{R}_K(u,t) := -\sum_{K<k\le t} \Delta_k(u,t).$$

Then, exactly,

$$M(t) = -\sum_{k=2}^{K} \Delta_k(u,t) + \mathcal{R}_K(u,t), \tag{10.6}$$

and $\mathcal{R}_K(u,t)$ uses only values $M(y)$ with

$$y \le \frac{\lambda u}{K+1}.$$

Equivalently, all dependence on values above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $\Delta_2, \dots, \Delta_K$. Moreover

$$\Delta_k(u,t) = \sum_{u/k<n\le t/k} \mu(n),$$

so the supporting real interval has length $(t-u)/k$ and contains $\lfloor t/k \rfloor - \lfloor u/k \rfloor$ integers.

Proof. Equation (10.6) is Proposition 10.5 with the terms $2 \le k \le K$ separated. If $k > K$, then $t/k \le \lambda u/(K+1)$ and $u/k \le \lambda u/(K+1)$. The last identity is the definition of the summatory function on the interval $(u/k, t/k]$. □

Corollary 10.7 (First channels)**.** [N-el] *For $K = 2$,*

$$M(t) = -\big(M(t/2) - M(u/2)\big) + \mathcal{R}_2(u,t),$$

where $\mathcal{R}_2$ uses only $y \le \lambda u/3$. With the convenient localized choice $\lambda = 5/4$, the only channel above $5u/12$ is

$$\Delta_2(u, u+r) = \sum_{u/2<n\le u/2+r/2} \mu(n), \qquad 0 < r \le u/4,$$

whose supporting interval has length at most $u/8$. In the limiting sense $\lambda \downarrow \lambda_*$, the successive remainder cutoffs are

$$\frac{\lambda_*}{2}u = 0.62445\ldots u, \quad \frac{\lambda_*}{3}u = 0.4162\ldots u, \quad \frac{\lambda_*}{4}u = 0.3122\ldots u, \quad \frac{\lambda_*}{5}u = 0.2497\ldots u.$$

Lemma 10.8 (Channel persistence extracted from a high excursion)**.** [N-el] *Let $S_r = \sigma_\nu M(u+r) > 0$ on $1 \le r \le h$ with $S_0 = 0$ and $|S_r - S_{r-1}| \le 1$. Fix $K \ge 2$ and write, using Proposition 10.6,*

$$S_r = \sum_{k=2}^{K} X_k(r) + Y_K(r), \quad X_k(r) := -\sigma_\nu \Delta_k(u, u+r), \quad Y_K(r) := \sigma_\nu \mathcal{R}_K(u, u+r).$$

If $H = \max_{r\le h} S_r$ and τ is a point with $S_\tau = H$, then among the last $\lfloor H/2 \rfloor + 1$ indices

$$\tau - \lfloor H/2 \rfloor, \dots, \tau$$

there is one fixed component among $X_2, \dots, X_K, Y_K$ which is at least $H/(2K)$ at no fewer than $\lfloor H/(2K) \rfloor$ indices.

Proof. Since $S_0 = 0$ and the path has unit steps, $\tau \ge H$ and $S_{\tau-j} \ge H - j \ge H/2$ for $0 \le j \le \lfloor H/2 \rfloor$. At every one of these indices the sum of the K displayed components is at least $H/2$, hence some component is at least $H/(2K)$. Pigeonholing the indices among the K components gives the claim. □

Remark 10.9 (What the persistence lemma adds). Lemma 10.8 converts a large value attained inside one excursion into a quantified persistence requirement: either one of the finitely many scaled interval sums $\Delta_2, \dots, \Delta_K$ repeatedly carries a positive fraction of the height, or the responsibility lies repeatedly in the deeper remainder, which is supported entirely below $\lambda u/(K+1)$. This avoids the iteration defect of Remark 10.22: no new point is claimed to be the start-window point of another excursion.

Corollary 10.10 (Channel descent with explicit loss)**.** [N-el] *Let u be a zero of M, $t > u$, $2 \le K \le u$, and $0 < \vartheta < 1$. Then either*

(i) *there are $2 \le k \le K$ and $y \in \{\lfloor u/k \rfloor, \lfloor t/k \rfloor\}$ with $|M(y)| \ge |M(t)|/(2K)$, hence*

$$\frac{|M(y)|}{y^\vartheta} \ge \frac{k^\vartheta}{2K} \cdot \frac{|M(t)|}{t^\vartheta}, \qquad y \le \left\lfloor \frac{t}{2} \right\rfloor;$$

or

(ii) *$|\mathcal{R}_K(u,t)| \ge |M(t)|/K$, where $\mathcal{R}_K$ depends only on $M(y)$ with $y \le \lfloor t/(K+1) \rfloor$.*

The statement holds at every $t > u$, not only in a localized window.

Proof. The proof of Proposition 10.6 applies verbatim to every $t > u$, with $\mathcal{R}_K$ supported on $y \le t/(K+1)$. By (10.6), one of the K terms $\Delta_2, \dots, \Delta_K, \mathcal{R}_K$ has modulus at least $|M(t)|/K$. If it is $\Delta_k = M(\lfloor t/k \rfloor) - M(\lfloor u/k \rfloor)$, one of its two values has modulus at least $|M(t)|/(2K)$. Since $k \le K \le u$, both candidate arguments are positive integers; hence $y \ge 1$. Moreover $y \le \lfloor t/k \rfloor \le \lfloor t/2 \rfloor$, and therefore

$$\frac{|M(y)|}{y^\vartheta} \ge \frac{|M(t)|}{2K} \left(\frac{t}{k}\right)^{-\vartheta} = \frac{k^\vartheta}{2K} \frac{|M(t)|}{t^\vartheta}.$$

□

Remark 10.11 (Channels do not iterate to a bound). Corollary 10.10 has no iteration gap: it applies at the new point y relative to the last zero of M before y. Its defect is the constant loss $k^\vartheta/(2K) < 1$ per step. If alternative (i) held at every step, a chain of length $m \asymp \log t$ would give only $|M(t)| \le (2K)^m\, O(1) = t^{O(\log K)}$, weaker than the trivial bound. For fixed K and $0 < \vartheta < 1$ the guaranteed factor can never reach 1, since

$$\frac{k^\vartheta}{2K} \le \frac{K^\vartheta}{2K} = \frac{1}{2K^{1-\vartheta}} < \frac{1}{2}.$$

Thus no iteration of the same pigeonhole estimate can remove the loss at an individual step. A closing argument must instead recover the loss collectively from the persistence supplied by Lemma 10.8, or prove cancellation in the deep remainder. Both require information beyond dependency alone.

Proposition 10.12 (Quantified persistence along channel descent)**.** [N] *Fix $K \ge 2$, $0 < \vartheta < 1$ and $\delta > 0$, and put*

$$c_{\vartheta,K} := \log\left(\frac{2K}{2^\vartheta}\right) > 0.$$

Suppose $t_0 \to \infty$ through points satisfying

$$Z_\vartheta(t_0) := \frac{|M(t_0)|}{t_0^\vartheta} \ge t_0^\delta.$$

Since $|M(t_0)| \le t_0$, any nonempty asymptotic regime of this form has $\delta \le 1 - \vartheta$.

At stage j, let u_j be the last zero of M before t_j. Continue the descent only while $u_j \ge K$ and alternative (i) *of Corollary 10.10 occurs; stop if either the deep-remainder alternative occurs or $u_j < K$. Whenever a descent step is available there are $2 \le k_j \le K$ and $t_{j+1} \le \lfloor t_j/2 \rfloor$ such that*

$$Z_\vartheta(t_{j+1}) \ge \frac{k_j^\vartheta}{2K} Z_\vartheta(t_j).$$

Consequently, after m successful steps,

$$Z_\vartheta(t_m) \geq Z_\vartheta(t_0) \prod_{j<m} \frac{k_j^\vartheta}{2K} \geq t_0^\delta e^{-mc_{\vartheta,K}}. \tag{10.7}$$

Set

$$m_* := \left\lfloor \frac{\delta \log t_0}{2c_{\vartheta,K}} \right\rfloor, \qquad \eta_{\vartheta,K,\delta} := \frac{\delta \log 2}{2c_{\vartheta,K}}.$$

Then, for all sufficiently large t_0, either a deep-remainder alternative occurs before step m_*, or the descent is valid through step m_* and there is a descendant satisfying

$$t_{m_*} \leq t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}, \qquad Z_\vartheta(t_{m_*}) \geq t_0^{\delta/2}.$$

Proof. Each successful application of Corollary 10.10 gives $t_{j+1} \leq \lfloor t_j/2 \rfloor$ and multiplies the normalized value by at least $2^\vartheta/(2K) = e^{-c_{\vartheta,K}}$, proving (10.7). Hence for every $m \leq m_*$ reached without a deep remainder,

$$Z_\vartheta(t_m) \geq t_0^{\delta/2}, \qquad t_m \leq t_0/2^m.$$

It remains to justify that the stopping condition $u_j < K$ cannot occur before m_*. Since M has infinitely many zeros, choose once and for all a zero $z_K \geq K$. If the last zero before t_j were $< K$, then necessarily $t_j < z_K$; otherwise z_K itself would be a preceding zero. Hence

$$Z_\vartheta(t_j) \leq z_K^{1-\vartheta},$$

a constant depending only on K and ϑ. This contradicts $Z_\vartheta(t_j) \geq t_0^{\delta/2}$ for large t_0. Therefore the only possible premature stop is the deep-remainder alternative. If it does not occur, the iteration reaches m_*. Finally $t_{m_*} \leq t_0 2^{-m_*} = t_0^{1-\eta_{\vartheta,K,\delta}+o(1)}$. □

Remark 10.13 (What remains after quantified persistence). Proposition 10.12 shows that a polynomial normalized violation cannot disappear immediately under channel descent. The pointwise loss accumulates, and bounding the deep remainder by absolute values loses a factor of the number of channels. A closing argument must therefore exploit the multi-index persistence of Lemma 10.8 collectively, or establish cancellation among the channels entering $\mathcal{R}_K$.

Corollary 10.14 (RH in the localized linear window)**.** [C] *RH holds if and only if, for every $\varepsilon > 0$,*

$$\left| \sum_{d \leq 5u/8} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right) \right| = \left| \sum_{2 \leq k \leq t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right] \right| \ll_\varepsilon t^{1/2+\varepsilon}$$

for every excursion starting at u and every

$$u < t \leq \min\left\{ x_1, \frac{5}{4}u \right\}.$$

Equivalently, in intra-excursion coordinates $t = u + r$ it is enough to control

$$1 \leq r \leq \min\{\ell - 1, u/4\}.$$

Proof. Use Corollary 10.3 with $\chi = 5/8$, together with Proposition 10.5. □

Proposition 10.15 (Kernel form of the recursion)**.** [N-el] *Let $M(u) = 0$ and let t be an integer with $u < t < 2u$. For $1 \leq y \leq \lfloor t/2 \rfloor$ put*

$$c_y(t,u) = \left(\left\lfloor \frac{t}{q} \right\rfloor - \left\lfloor \frac{t}{y+1} \right\rfloor \right) - \left(\left\lfloor \frac{u}{q} \right\rfloor - \left\lfloor \frac{u}{y+1} \right\rfloor \right).$$

Then:

(i) $M(t) = -\sum_{1 \leq y \leq \lfloor t/2 \rfloor} c_y(t,u) M(y)$;

(ii) $c_y(t,u) = \dfrac{t-u}{y(y+1)} + e_y$ *with* $|e_y| < 2$*; for* $y > \sqrt{t}$ *one has* $c_y \in \{-1,0,1\}$*, and* $c_y \neq 0$ *only if* $y = \lfloor t/k \rfloor$ *or* $y = \lfloor u/k \rfloor$ *for some* $2 \le k < \sqrt{t}$ *(the value* $k=1$ *gives* $y \in \{t,u\} > \lfloor t/2 \rfloor$*);*
(iii) $c_1(t,u) \ge \frac{1}{2}(t-u-1)$*;*
(iv) $M(t) = -(t-u)\,\Upsilon(\sqrt{t}) + \mathrm{Err}(t,u)$*, where* Υ *is the coefficient* (10.22) *and*

$$|\mathrm{Err}(t,u)| \le 2\sum_{y\le\sqrt{t}} |M(y)| + \sum_{2\le k<\sqrt{t}} \Big(\Big|M\Big(\frac{t}{k}\Big)\Big| + \Big|M\Big(\frac{u}{k}\Big)\Big| \Big).$$

Proof. (i) In (10.5) group the indices $k \ge 2$ by the value $y = \lfloor t/k \rfloor$, respectively $y = \lfloor u/k \rfloor$. The number of $k \ge 1$ with $\lfloor t/k \rfloor = y$ is $\lfloor t/y \rfloor - \lfloor t/(y+1) \rfloor$, and for $y \le \lfloor t/2 \rfloor$ none of them is $k = 1$. The same holds for u, because $k = 1$ gives $y = u > \lfloor t/2 \rfloor$. (ii) Each bracket equals $x/(y(y+1))$ minus a difference of two fractional parts, so it differs from $x/(y(y+1))$ by less than 1. For $y > \sqrt{t} \ge \sqrt{u}$ both $t/y - t/(y+1)$ and $u/y - u/(y+1)$ are < 1, so each bracket is 0 or 1, and it is 1 only if $y = \lfloor x/k \rfloor$ with $k \le x/y < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2 \rfloor - \lfloor u/2 \rfloor)$ and $\lfloor t/2 \rfloor - \lfloor u/2 \rfloor \le \lceil (t-u)/2 \rceil$. (iv) Insert (ii) into (i) for $y \le \sqrt{t}$; the smooth part is $(t-u)\sum_{y\le\sqrt{t}} M(y)/(y(y+1)) = (t-u)\Upsilon(\sqrt{t})$. For $y > \sqrt{t}$ use the last assertion of (ii). □

Corollary 10.16 (Loss of the pointwise induction)**.** [N-el] *Let* $0 < \vartheta < 1$ *and suppose* $|M(y)| \le Cq^{\vartheta}$ *for all* $y \le \lfloor t/2 \rfloor$*. Then, for* $u < t < 2u$ *with* $M(u) = 0$*,*

$$\big|M(t) + (t-u)\Upsilon(\sqrt{t})\big| \ll_{\vartheta} C\,t^{(1+\vartheta)/2}, \tag{10.8}$$

while the direct absolute-value estimate for the full kernel is, in general, of linear size and yields at best $|M(t)| \ll_{\vartheta} C\,t$ *from this information alone. For* $\vartheta = \frac{1}{2} + \varepsilon$ *the pointwise induction therefore loses* $t^{1/2-\varepsilon}$*, and* $t^{1/4-\varepsilon/2}$ *after the smooth component* $-(t-u)\Upsilon(\sqrt{t})$ *is extracted.*

Proof. By Proposition 10.15(iv), $|\mathrm{Err}| \le 2C\sum_{y\le\sqrt{t}} y^{\vartheta} + 2Ct^{\vartheta}\sum_{2\le k<\sqrt{t}} k^{-\vartheta} \ll_{\vartheta} Ct^{(1+\vartheta)/2}$. The full kernel satisfies $\sum_y |c_y| y^{\vartheta} \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_y |c_y| y^{\vartheta} \ll_{\vartheta} t$ by (ii). □

The smooth component is not small for termwise reasons: by (10.24),

$$\Upsilon(\sqrt{t}) = -\sum_{n>\sqrt{t}} \frac{M(n)}{n(n+1)},$$

so its size is governed by the global identity $\sum_{n\ge1} M(n)/(n(n+1)) = 0$, i.e. by the pole of ζ at $s = 1$ (Remark 10.52), and involves values of M beyond $t/2$. Even under RH the tail gives only $(t-u)\Upsilon(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $\mathrm{Err}(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

Remark 10.17 (Localized intra-excursion system). Let $I = (u, x_1]$ be an excursion of length $\ell = x_1 - u$ and orientation σ_ν, and put

$$S_r := \sigma_\nu M(u+r) > 0 \qquad (1 \le r < \ell), \qquad S_0 = S_\ell = 0.$$

For a fixed $\lambda \in (\lambda_*, 2)$ set

$$h_\lambda(u) := \min\{\ell - 1, (\lambda - 1)u\}.$$

Then the entire RH-relevant initial segment satisfies simultaneously

$$S_r = -\sigma_\nu \sum_{y\le\lambda u/2} c_y(u+r,u) M(y) > 0, \qquad 1 \le r \le h_\lambda(u),$$

and $|S_r - S_{r-1}| \le 1$. For $\lambda = 5/4$ this is the segment $r \le \min\{\ell - 1, u/4\}$ and all dependencies lie below $5u/8$.

Definition 10.18 (Relaxed divisor-sum window class)**.** Fix $u \geq 2$ and $1 \leq h < u$. For $\mathbf{a} = (a_d)_{d \leq u} \in \{-1, 0, 1\}^u$ with $a_1 = 1$ and $\sum_{d \leq u} a_d = 0$, put

$$D_{\mathbf{a}}(n) := \sum_{\substack{d \mid n \\ d \leq u}} a_d, \qquad u < n \leq u + h.$$

We call $\mathbf{a}$ window-admissible with orientation $\sigma_\nu \in \{-1, 1\}$ if $D_{\mathbf{a}}(n) \in \{-1, 0, 1\}$ throughout the window and

$$\sigma_\nu \sum_{u < n \leq u + r} D_{\mathbf{a}}(n) < 0 \qquad (1 \leq r \leq h).$$

The induced oriented path is $S_r = -\sigma_\nu \sum_{u < n \leq u + r} D_{\mathbf{a}}(n) > 0$. For the genuine Möbius prefix, $a_d = \mu(d)$, these are exactly the increment and excursion conditions coming from $\mu * 1 = \delta_1$.

Proposition 10.19 (Ballistic counterexample to the relaxed window class)**.** [N-el] *The relaxed class of Definition 10.18 admits ballistic paths. Namely, take*

$$a_1 = 1, \qquad a_u = -1, \qquad a_d = 0 \quad (1 < d < u).$$

Then for every $u < n < 2u$ one has $D_{\mathbf{a}}(n) = 1$, and with $\sigma_\nu = -1$,

$$S_r = r \qquad (1 \leq r < u).$$

If $m_y := \sum_{d \leq y} a_d$, then $m_y = 1$ for every $y < u$. Hence for any $0 < \vartheta < 1$, any fixed $1 < \lambda < 2$, and $h = \lfloor (\lambda - 1)u \rfloor$,

$$\sup_{y \leq \lambda u/2} \frac{|m_y|}{y^\vartheta} = 1, \qquad \frac{S_h}{(u + h)^\vartheta} \asymp_\lambda u^{1-\vartheta} \longrightarrow \infty.$$

Thus a maximum principle comparing the normalized localized-window height to the normalized history is false on this relaxed class.

Proof. For $u < n < 2u$, the integer u does not divide n. Hence among the two nonzero coordinates of $\mathbf{a}$ only $d = 1$ contributes to $D_{\mathbf{a}}(n)$, so $D_{\mathbf{a}}(n) = 1$. The remaining assertions are immediate. □

Remark 10.20 (What the relaxation loses). The counterexample violates the internal divisor identities of the Möbius prefix. Already for every prime $p < u$ the genuine identity requires $a_1 + a_p = 0$, whereas the ballistic vector above has $a_p = 0$. Consequently any successful localized argument of this type must use some of the relations

$$\sum_{d \mid n} a_d = 0 \qquad (2 \leq n \leq u),$$

not merely the future-window constraints.

Remark 10.21 (The prefix correlation identity). The complete-excursion bridge identity must not be applied to a truncated localized window. For $1 \leq h < \ell$, put $x_j = \sigma_\nu \mu(u + j)$, $S_r = \sum_{j \leq r} x_j$, and $C_g^{\langle h \rangle} = \sum_{j=1}^{h-g} x_j x_{j+g}$. Lemma 5.1 applied to the prefix of length h gives exactly

$$\sum_{r=1}^{h-1} S_r^2 + \sum_{g=1}^{h-1} g C_g^{\langle h \rangle} = S_h \sum_{r=1}^{h-1} S_r.$$

The right-hand side vanishes only for a complete bridge. Thus $Q_\nu = -\sum_a a C_a^{(\nu)}$ cannot be used as an identity internal to a truncated horizon unless the excursion actually returns there. In particular, passing from a large localized value to the energy of the whole excursion mixes the retained window with the part deliberately discarded by the horizon theorem.

Remark 10.22 (The one-step iteration gap). Even a one-step implication sending a large value in the localized window to a large value at some $y \le \lambda u/2$ would not automatically iterate. The point y need not lie in the localized window of the excursion that contains y; it may lie beyond $\lambda u'$ relative to that excursion's start u'. Sparsity of the family of λ-long excursions does not repair a single descent chain. Iteration therefore requires either a principle valid for all t, or a mechanism that places the new point inside an admissible window at every step.

Proposition 10.23 (Termwise absolute-value contraction gives no nontrivial gain)**.** [N-el] *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_y = \sum_{r \le h} w_r c_y(u + r, u)$, so that $\sum_r w_r M(u + r) = -\sum_q K_q M(y)$. If, for some $0 < \vartheta < 1$, $\alpha \ge 0$ and $C_\alpha > 0$,*

$$\sum_y |K_y| y^\vartheta \le C_\alpha \sum_{r \le h} w_r (u + r)^\alpha,$$

then

$$\sum_{r \le h} w_r r \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u + r)^\alpha, \qquad \sum_{r \le h} w_r |M(u + r)| \le (1 + 2C_\alpha) \sum_{r \le h} w_r (u + r)^\alpha.$$

Thus the contraction hypothesis can hold only on windows where the trivial bound $|M(u + r)| \le r$ already gives the conclusion.

Proof. Proposition 10.15(iii) gives $c_1(u + r, u) \ge (r - 1)/2 \ge 0$, hence $\frac{1}{2} \sum_r w_r (r - 1) \le K_1 \le C_\alpha \sum_r w_r (u + r)^\alpha$. Since $M(u) = 0$ and M has unit steps, $|M(u + r)| \le r$; since $(u + r)^\alpha \ge 1$, $\sum w_r \le \sum w_r (u + r)^\alpha$, and both inequalities follow. □

Remark 10.24 (No strict coercivity from the first coefficient). Writing $t = u + r$ gives the exact formula

$$c_1(u + r, u) = \begin{cases} \lceil r/2 \rceil, & u \text{ even}, \\ \lfloor r/2 \rfloor, & u \text{ odd}. \end{cases}$$

In particular $c_1(u + 1, u) = 0$ when u is odd. The preceding proposition therefore rules out only this particular termwise absolute-value mechanism; it does not rule out a global signed contraction or cancellation among the coefficients K_y.

Definition 10.25 (Multiplicative layers of the prefix)**.** Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity. For $P \ge 2$ and $k \ge 1$, a vector $(a_d)_{d \le P}$ satisfies the first k Möbius layers if $a_1 = 1$ and

$$\sum_{d | n} a_d = 0 \qquad (2 \le n \le P,\ \Omega(n) \le k).$$

Thus layer 1 imposes the identities at primes, layer 2 adds prime squares and products of two primes, and so on.

Proposition 10.26 (One-layer propagation into the localized future)**.** [N-el] *Suppose $(a_d)_{d \le P}$ satisfies the first k Möbius layers. Then*

$$a_d = \mu(d) \qquad (d \le P,\ \Omega(d) \le k).$$

Moreover, for every integer n with

$$P < n \le 2P, \qquad \Omega(n) \le k + 1,$$

one has

$$-\sum_{\substack{d | n \\ d \le P}} a_d = \mu(n).$$

Thus the first k multiplicative layers of the prefix determine the future increments at every integer n in the stated window $P < n \le 2P$ with $\Omega(n) \le k + 1$.

Proof. The first assertion follows recursively from $a_n = -\sum_{d|n,\ d<n} a_d$: every proper divisor of an integer with $\Omega(n) \le k$ has smaller Ω. For the second, every proper divisor of $n \le 2P$ is at most $n/2 \le P$ and has $\Omega(d) \le k$. Hence the displayed sum equals $-\sum_{d|n,\ d<n} \mu(d) = \mu(n)$. □

Remark 10.27 (Multiplicative-layer depth as a quantitative invariant). Fix $\chi \in (\chi_*, 1)$ and the localized window $u < n \le 2\chi u$. Let $k = k(u)$ be a depth up to which the internal Möbius divisor identities are imposed below $P = \chi u$. What growth of $k(u)$, if any, suffices to combine those identities with $M(u) = 0$ and the unit-step and excursion-sign constraints so as to force $\max_{u<t\le 2\chi u} |M(t)| \ll_\varepsilon u^{1/2+\varepsilon}$? A bounded depth would give a compressed arithmetic mechanism; imposing all layers reconstructs the Möbius prefix. Proposition 10.26 makes the depth a quantifiable intermediate invariant; the results below treat only fixed k.

Proposition 10.28 (Layer sums in the localized window)**.** [L] *Fix $1 < \lambda < 2$ and an integer $j \ge 1$ before letting $u \to \infty$. Then*

$$\sum_{\substack{u<n\le \lambda u \\ \Omega(n)=j}} \mu(n) = (-1)^j\,(1+o(1))\,\frac{(\lambda-1)\,u\,(\log\log u)^{j-1}}{(j-1)!\,\log u}. \tag{10.9}$$

Proof. If $\Omega(n) = j$ and $\mu(n) \ne 0$, then n is squarefree with exactly j prime factors and $\mu(n) = (-1)^j$. By Landau's theorem (see [45, Ch. 7]), the number $\pi_j^\flat(x)$ of such $n \le x$ satisfies $\pi_j^\flat(x) \sim f_j(x) := x(\log\log x)^{j-1}/((j-1)!\log x)$. For fixed j and fixed $\lambda > 1$,

$$\frac{f_j(\lambda u)}{f_j(u)} = \lambda\,\frac{(\log\log(\lambda u))^{j-1}}{(\log\log u)^{j-1}}\,\frac{\log u}{\log(\lambda u)} \longrightarrow \lambda.$$

Hence $f_j(\lambda u) - f_j(u) = (\lambda - 1 + o(1))f_j(u)$, as claimed. □

Corollary 10.29 (Bounded layer depth)**.** [N] *Fix $k \ge 1$, $\chi \in (\chi_*, 1)$, $P = \chi u$, and the window $u < n \le 2\chi u$.*

(i) *The increments fixed by the first k layers (Proposition 10.26), namely those with $\Omega(n) \le k+1$, form a set of $o(u)$ integers of the window.*

(ii) *Their summed contribution over the window is*

$$\sum_{\substack{u<n\le 2\chi u \\ \Omega(n)\le k+1}} \mu(n) = (-1)^{k+1}(1+o(1))\frac{(2\chi-1)\,u\,(\log\log u)^k}{k!\,\log u},$$

so, for every fixed $\varepsilon > 0$ and all sufficiently large u, it exceeds $u^{1-\varepsilon}$ in modulus and has the fixed sign $(-1)^{k+1}$.

(iii) *Let $a_d = \mu(d)$ for $\Omega(d) \le k$ and $a_d = 0$ otherwise. For squarefree n in the window with $\omega(n) = w \ge k+2$,*

$$D_{\mathbf{a}}(n) = \sum_{\substack{d|n \\ d\le P}} a_d = \sum_{j=0}^{k} (-1)^j \binom{w}{j} = (-1)^k \binom{w-1}{k},$$

which has modulus at least $k+1 \ge 2$. Thus the zero extension of a k-layer prefix violates the unit-step constraint at every such n, i.e. at all but $o(u)$ squarefree integers of the window.

Proof. (ii) follows from Proposition 10.28 summed over $j \le k+1$, the term $j = k+1$ dominating. For (i), Landau's asymptotic $\#\{n \le x : \Omega(n) = j\} \sim x(\log\log x)^{j-1}/((j-1)!\log x)$ for fixed j [45, Ch. 7] (all integers, not only squarefree ones) shows that the window contains $o(u)$ integers with $\Omega(n) \le k+1$. For (iii), all divisors of n other than n are $\le n/2 \le P$, and n itself has $\Omega(n) = w > k$, so $D_{\mathbf{a}}(n)$ is the truncated alternating binomial sum, evaluated by induction on k

via $\binom{w-1}{k} = \binom{w}{k} - \binom{w-1}{k-1}$. The proportion of squarefree $n \le x$ with $\omega(n) \le k+1$ tends to 0 by Proposition 10.28. □

Remark 10.30 (What fixed layer depth proves). For every fixed k, Corollary 10.29 shows that the first k layers determine only a density-zero subset of the localized window, while the signed sum of the determined increments has modulus $(1+o(1))(2\chi-1)u(\log\log u)^k/(k!\log u)$, larger than $u^{1-\varepsilon}$ for every fixed $\varepsilon > 0$, once u is sufficiently large. Hence, if the full sum on the window is $O(u^{1/2+\varepsilon})$, substantial cancellation must occur between the increments determined by any fixed set of initial layers and the remaining ones. The results proved here do not determine how the required depth grows with u; in particular they do not show that $\asymp \log\log u$ layers (the typical size of Ω) are necessary or sufficient, which would require estimates uniform in $k = k(u)$. Part (iii) shows that the unit-step constraint is far from vacuous on a k-layer relaxation: it forbids the naive completion at almost every integer of the window, so the free coordinates a_d with $\Omega(d) > k$ are strongly constrained.

10.2. **The square range as an exact bilinear form of the prefix.**

Proposition 10.31 (Bilinear square-range identity)**.** [L] *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor. \tag{10.10}$$

In particular, if $M(u) = 0$ and $u < t \le u^2$,

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{10.11}$$

Proof. Let $\mu_u = \mu\mathbf{1}_{[1,u]}$. Each factor of $(\mu-\mu_u)*(\mu-\mu_u)$ is supported on integers $\ge u+1$, so the convolution, and its convolution with 1, vanish on $n < (u+1)^2$, in particular on $n \le u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \le u^2$; sum over $n \le t$. □

Remark 10.32 (Higher-order determinacy). For every $k \ge 1$ and $n \le u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1}\binom{k}{j}\big(\mu_u^{*j} * 1^{*(j-1)}\big)(n). \tag{10.12}$$

Indeed, $(\mu-\mu_u)^{*k}$ is supported on integers at least $(u+1)^k$, so after convolution with $1^{*(k-1)}$ it still vanishes for $n < (u+1)^k$, in particular for $n \le u^k$. Expanding gives

$$0 = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \mu^{*(k-j)} * \mu_u^{*j} * 1^{*(k-1)}.$$

For $j = 0$, $\mu^{*k} * 1^{*(k-1)} = \mu$. For $j \ge 1$, pair $k-j$ factors of μ with $k-j$ of the factors 1 to obtain $\mu^{*(k-j)} * 1^{*(k-j)} = \delta$; the remaining $j-1$ factors of 1 produce the term in (10.12). Rearranging proves the formula. Thus determination by the prefix is not special to u^2. What is special to $t \le u^2$ is the order-two form (10.11), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj)\rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [35]; summed forms of this kind are used in the fast computation of M [36].

Remark 10.33 (Square-excursion bridges). Let $u = x_{0,\nu}$ and $x_{1,\nu} \le u^2$. Then $\mathcal{B}_u(u) = \mathcal{B}_u(x_{1,\nu}) = 0$ and $-\sigma_\nu \mathcal{B}_u(t) > 0$ for $u < t < x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj)\rfloor$ and these bridge and sign constraints be combined to prove the bound

$$\max_{u<t\le x_{1,\nu}} |\mathcal{B}_u(t)| \ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon}$$

uniformly over square-ordinary excursions? By Corollary 9.12 this uniform bound is equivalent to RH. By Corollary 10.14 it suffices to consider $t \le \min\{x_{1,\nu}, \frac{5}{4}u\}$, where $\mathcal{B}_u(t)$ reduces to the localized linear form and uses only $\mu(d)$ with $d \le 5u/8$.

10.3. **Trial-division horizon of an excursion.**

Lemma 10.34 (Square horizon)**.** [L] *Let $u \ge 2$. Trial division by the primes $p \le u$ alone determines $\mu(n)$ for every $n \le u^2$.*

Proof. Remove the prime factors $p \le u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \le u^2$ has all prime factors $> u$, so it is 1 or a single prime. □

Remark 10.35 (Determinacy is not information). Primes $> u$ are not logically new information, and determination by a finite prefix is not special to u^2: (10.12) propagates a prefix to every power range, and (10.4) determines M from $M(1)$. The ranges $(u, 2u]$ and $(u, u^2]$ are special only in the concrete sense that the exact summatory formula is of order one ((10.2)) or two ((10.11)) in the prefix, and that one-stage trial division by primes $\le u$ suffices on $(u, u^2]$.

Definition 10.36 (Smooth and rough parts)**.** For $y > 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\le y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 10.37. [L] *$\mu(n) = \mu(s_y(n))\,\mu(r_y(n))$. If $n < y^k$ then $\Omega(r_y(n)) \le k-1$, and if $n \le y^2$ then $r_y(n)$ is 1 or a prime.*

Proof. The two parts are coprime. If $\Omega(r) \ge 1$ then $y^{\Omega(r)} < r \le n$. □

Remark 10.38 (The horizon u^2 is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, below u_ν. The regime $t > u_\nu^2$ of Section 10.4 lies beyond v_ν for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 10.5 gives the local version inside an excursion.

10.4. **Large-prime decomposition at level U^2.** The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [45, 53].

Definition 10.39 (Smooth baseline and large-prime part)**.** For $U \ge 2$ and $t \ge 1$ put

$$M_U^{\mathrm{sm}}(t) := \sum_{\substack{n \le t \\ P^+(n) \le U^2}} \mu(n), \qquad M_U^{\mathrm{lp}}(t) := \sum_{\substack{n \le t \\ P^+(n) > U^2}} \mu(n), \qquad M(t) = M_U^{\mathrm{sm}}(t) + M_U^{\mathrm{lp}}(t). \tag{10.13}$$

Proposition 10.40 (Single-prime copy)**.** [L] *For a prime p and $t < p^2$,*

$$\mathcal{D}_p(t) := \sum_{a \le t/p} \mu(ap) = -M(t/p). \tag{10.14}$$

Proof. $a \le t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 10.41 (Sign content of one prime cone)**.** [N-el] *Let $\mathcal{S}(y) = \sum_{a \le y} \mu(a)^2$. Among the numbers ap with $a \le y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{10.15}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proof. $\mu(ap) = -\mu(a)$, and $\#\{a \le y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

Proposition 10.42 (First-order window)**.** [L] *If $U^2 < t < U^4$, no $n \le t$ has two prime factors $> U^2$ (with multiplicity), and*

$$M_U^{\mathrm{lp}}(t) = -\sum_{U^2<p\le t} M(t/p). \tag{10.16}$$

Proof. Two such factors would give $n > U^4 > t$. Hence $M_U^{\mathrm{lp}}(t) = \sum_{U^2<p\le t} \mathcal{D}_p(t)$, and $t < U^4 < p^2$; apply (10.14). □

Proposition 10.43 (Prime-count form)**.** [N-el] *For $U^2 < t < U^4$ and $W := t/U^2$,*

$$M_U^{\mathrm{lp}}(t) = -\sum_{a\le W} \mu(a)\big(\pi(t/a) - \pi(U^2)\big), \qquad M(t) = M_U^{\mathrm{sm}}(t) - \sum_{a\le W} \mu(a)\pi(t/a) + M(W)\pi(U^2). \tag{10.17}$$

Proof. Expand (10.16) as $-\sum_{U^2<p\le t}\sum_{a\le t/p}\mu(a)$. This is a finite sum, so reversing the order is purely combinatorial. For fixed a the prime runs over $U^2 < p \le t/a$, which is non-empty only if $a < W$. The displayed sum may harmlessly be written with $a \le W$: if W is an integer, the endpoint $a = W$ contributes $\pi(U^2) - \pi(U^2) = 0$. □

Thus the only input from primes $> U^2$ is the value of π at the points t/a, $a \le W$, weighted by $\mu(a)$ at the small scale $W < U^2$.

Definition 10.44 (Copy multiplicities)**.** For $U^2 < t < U^4$, $W = t/U^2$ and $1 \le y \le \lfloor W \rfloor$ put

$$N_y(t,U) = \#\Big\{p:\ \max\big(U^2, \tfrac{t}{y+1}\big) < p \le \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad M_U^{\mathrm{lp}}(t) = -\sum_{1\le y\le \lfloor W\rfloor} N_y(t,U) M(y). \tag{10.18}$$

Remark 10.45 (Asymptotics of the multiplicities). For fixed $W > 1$ and $U \to \infty$,

$$N_y \sim \frac{WU^2}{2\log U}\cdot\frac{1}{y(y+1)} \quad (1 \le y \le \lfloor W\rfloor - 1), \qquad N_{\lfloor W\rfloor} \sim \frac{U^2}{2\log U}\Big(\frac{W}{\lfloor W\rfloor} - 1\Big) \quad (W \notin \mathbb{Z}), \tag{10.19}$$

and $N_W = 0$ if $W \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor W\rfloor + 1) < U^2$.

Definition 10.46 (Coherent coefficient)**.**

$$\mathcal{C}(W) := \int_1^W \frac{M(w)}{w^2}\,dw = \sum_{a\le W}\mu(a)\Big(\frac{1}{a} - \frac{1}{W}\Big) = \sum_{a\le W}\frac{\mu(a)}{a} - \frac{M(W)}{W}. \tag{10.20}$$

Proposition 10.47 (Fixed W)**.** [N-el] *For fixed $W > 1$, as $U \to \infty$,*

$$M_U^{\mathrm{lp}}(WU^2) = -\frac{WU^2}{2\log U}\big(\mathcal{C}(W) + o(1)\big). \tag{10.21}$$

Proof. In (10.17) the sum is finite; for $a < W$ the prime number theorem gives $\pi(WU^2/a) - \pi(U^2) \sim (W/a - 1)U^2/(2\log U)$, and $\sum_{a\le W}\mu(a)(W/a - 1) = W\,\mathcal{C}(W)$. □

Lemma 10.48 (The multiplicity-weighted coefficient)**.** [N-el] *The coefficient*

$$\Upsilon(W) := \sum_{1\le y\le W}\frac{M(y)}{y(y+1)} = \sum_{n\le W}\frac{\mu(n)}{n} - \frac{M(W)}{\lfloor W\rfloor + 1} \tag{10.22}$$

satisfies

$$\Upsilon(W) = \mathcal{C}\big(\lfloor W\rfloor + 1\big), \qquad \mathcal{C}(W) = \Upsilon(W) - M(W)\Big(\frac{1}{W} - \frac{1}{\lfloor W\rfloor + 1}\Big), \tag{10.23}$$

and

$$\mathcal{C}(W) = -\int_W^\infty \frac{M(w)}{w^2}\,dw, \qquad \Upsilon(W) = -\sum_{n>W}\frac{M(n)}{n(n+1)}. \tag{10.24}$$

Proof. M is constant on $[y, y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (10.23). For $s > 1$, (8.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s)) \to 0$ as $s \downarrow 1$. For $s \in [1, 1+\delta]$ the bound $M(w) \ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw = 0$. □

Remark 10.49 ($\mathcal{C}$ versus Υ). Summing (10.19) against $M(y)$ suggests the coefficient $\Upsilon(W)$, but the truncated top multiplicity replaces it by $\mathcal{C}(W)$; by (10.23) the two agree exactly when $M(\lfloor W \rfloor) = 0$. For $1 < W < 2$ one has $M_U^{\mathrm{lp}} = -(\pi(t) - \pi(U^2)) \sim -(W-1)U^2/(2\log U)$, i.e. $\mathcal{C}(W) = 1 - 1/W$, whereas $\Upsilon(W) = 1/2$.

Corollary 10.50 (Decay of the coefficients)**.** [C] *By* (10.24) *and* (7.2)*,*

$$\mathcal{C}(W),\ \Upsilon(W) \ll \exp\bigl(-c_1(\log W)^{3/5}(\log\log W)^{-1/5}\bigr). \tag{10.25}$$

Proposition 10.51 (Asymptotic size of the smooth and large-prime components)**.** [N]

(i) *Suppose $W > 1$ is fixed and $\mathcal{C}(W) \neq 0$. Then*

$$M_U^{\mathrm{lp}}(WU^2) \sim -\frac{WU^2}{2\log U}\mathcal{C}(W),$$
$$M_U^{\mathrm{sm}}(WU^2) \sim +\frac{WU^2}{2\log U}\mathcal{C}(W). \tag{10.26}$$

For $1 < W < 2$ one has $\mathcal{C}(W) = 1 - 1/W > 0$, so the hypothesis holds.

(ii) *There is $C > 0$ such that, as $t \to \infty$, uniformly for $U^2 < t < U^4$ with $W = t/U^2 \geq \exp(C(\log\log t)^2)$,*

$$M_U^{\mathrm{lp}}(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad M_U^{\mathrm{sm}}(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{10.27}$$

In particular, in both regimes $|M_U^{\mathrm{lp}}(t)|$ and $|M_U^{\mathrm{sm}}(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon < 1/2$ and all large t, and only their sum M can be compared with the scale $t^{1/2+\varepsilon}$.

Proof. We use the classical form [45]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O\bigl(ye^{-c\sqrt{\log y}}\bigr), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a \leq y}\frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{10.28}$$

and the series $\sum_{n\geq 1}\mu(n)\log n/n = -1$ converges conditionally. This follows, for example, from the displayed bound on $\widetilde{m}$ and partial summation (and agrees with the limit at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Bigl(\widetilde{m}(Y)\log Y - \sum_{a\leq Y}\frac{\mu(a)\log a}{a}\Bigr) = 1. \tag{10.29}$$

(i) Proposition 10.47, and $M(t) = o(t/\log t)$. For $1 < W < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < U^4$, $W < \sqrt{t}$, so $\log(t/a) \geq \frac{1}{2}\log t$ for $a \leq W$. By (10.17),

$$M_U^{\mathrm{lp}}(t) = -\sum_{a\leq W}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\leq W}\mu(a)\bigl(\pi(t/a) - \mathrm{li}(t/a)\bigr) + M(W)\pi(U^2). \tag{10.30}$$

The second term is $\ll t(1+\log W)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac{1}{4}\log t < \log U < \frac{1}{2}\log t$, so

$$|M(W)|\pi(U^2) \ll We^{-c\sqrt{\log W}}\frac{U^2}{\log U} \ll e^{-c\sqrt{\log W}}\frac{t}{\log t} = o(t/\log^2 t)$$

once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\le W} \mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(W)F(W) - \int_1^W \widetilde{m}(w)F'(w)\,dw, \tag{10.31}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big).$$

The boundary term is $\ll e^{-c\sqrt{\log W}} t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^W \widetilde{m}(w) g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1,4].$$

The required uniformity in (U, W) follows by an explicit ε–W argument. Given $\varepsilon_1 > 0$, choose W_0 so that

$$\int_{W_0}^{\infty} \frac{|\widetilde{m}(w)|}{w}\,dw < \varepsilon_1.$$

For all sufficiently large t, the lower bound on W gives $W > W_0$. On $[1, W_0]$,

$$\sup_{1\le w\le W_0} |g_t(w) - 1| \ll \frac{\log W_0}{\log t} \to 0,$$

while on $[W_0, W]$ one has $1 \le g_t(w) \le 4$. The omitted interval (W, ∞) is also contained in the chosen tail. Therefore

$$\int_1^W \widetilde{m}(w) g_t(w)\frac{dw}{w} = \int_1^{\infty} \widetilde{m}(w)\frac{dw}{w} + o(1) = 1 + o(1),$$

uniformly over all admissible pairs (U, W). Hence $M_U^{\mathrm{lp}}(t) = (1+o(1))t/\log^2 t$, and $M_U^{\mathrm{sm}} = M - M_U^{\mathrm{lp}}$ with $M(t) = o(t/\log^2 t)$. □

Remark 10.52 (Why $t/\log^2 t$). $\mathcal{C}(W)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 10.50. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and does not decay in W.

Remark 10.53 (First copy transitions). For $1 < W < 2$ every copy has length 1 and $M_U^{\mathrm{lp}}(t) = -(\pi(t) - \pi(U^2))$. For $2 < W < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < W < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

10.5. **Local large-prime decomposition of an excursion.**

Proposition 10.54 (Local decomposition)**.** [N-el] *Let $I_\nu = (u, v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}} \mu(n) - \sum_{a<y} \mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{10.32}$$

and the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le \sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{10.33}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a, p) \mapsto ap$ is a bijection onto these n. □

Remark 10.55 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Huxley proved the prime number theorem uniformly in intervals of length $h \geq x^{7/12+\varepsilon}$ [38]; the zero-density estimates of Guth–Maynard give intervals of length $x^{17/30+o(1)}$ [34]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\leq x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [44]. The cited results do not provide, below these ranges, an asymptotic for Π_a uniform in a.

Lemma 10.56 (Inverse logarithms). [C] *Let C be as in Proposition 10.51. Uniformly for*

$$\exp\bigl(C(\log\log x)^2\bigr) \leq z \leq \sqrt{x},$$

$$\sum_{a\leq z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{10.34}$$

Proof. Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw.$$

Since $z \leq \sqrt{x}$, the boundary term is uniformly $O(e^{-c\sqrt{\log z}}/\log x)$, hence $o(1/\log^2 x)$. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$ with $1 \leq g_x(w) \leq 4$, extend $g_x \mathbf{1}_{\{w\leq z\}}$ by zero, and apply dominated convergence with (10.29). □

Heuristic 10.57 (Opposite drifts). Let $u \geq \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}} \mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 10.56. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of σ_ν. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu,$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (10.33) is a balance between two opposite drifts: the local analogue of Proposition 10.51.

11. Two-point correlations and the energy of excursions

For integers $0 \leq u < X$ and $1 \leq h < X - u$ put

$$C_h^{(u,X]} := \sum_{u<n\leq X-h} \mu(n)\mu(n+h), \qquad C_h^{[X]} := C_h^{(0,X]}.$$

For an excursion $I_\nu = (x_{0,\nu}, x_{1,\nu}]$ one has $C_h^{(\nu)} = C_h^{(x_{0,\nu},x_{1,\nu}]}$, since $\sigma_\nu^2 = 1$. Chowla's conjecture asserts $C_h^{[X]} = o(X)$ for each fixed $h \geq 1$. Matomäki, Radziwiłł and Tao [42] prove, for $K = K(X) \to \infty$, $K \leq X$,

$$\sum_{h\leq K} \left| \sum_{n\leq X} \mu(n)\mu(n+h) \right| = o(KX).$$

Since replacing the upper limit X by $X - h$ changes the h-th sum by at most h, we shall use the triangular consequence

$$\sum_{h\leq K} \bigl|C_h^{[X]}\bigr| = o(KX) \qquad (K \to \infty,\ K = o(X)). \tag{11.1}$$

Their stronger averaged Elliott form permits translated shifts of size $O(X)$ and covers μ.

11.1. **Signed lag averages and the weighted RH criterion.**

Remark 11.1 (Unweighted signed averages are degenerate). By Lemma 5.3, every excursion satisfies $\sum_h C_h^{(\nu)} = -q_\nu/2$, with $q_\nu \le \ell_\nu$. Hence $\ell_\nu^{-2}\sum_h C_h^{(\nu)} = O(\ell_\nu^{-1})$ for purely algebraic reasons: a signed average over lags of the correlations inside a bridge is small without any pseudorandomness of μ. Statements of Chowla type inside an excursion must therefore carry absolute values inside the lag average, or impose a genuinely weighted condition.

The weighted sum behaves differently. By (5.5), $Q_\nu = -\sum_h hC_h^{(\nu)}$: the weight h records at which lags the correlation responsible for the excursion energy sits. The same identity holds globally at every zero of M, and the cross terms between different excursions cancel exactly.

Proposition 11.2 (Weighted correlations at zeros of M)**.** [N-el] *Let $v \ge 2$ be a zero of M. Then*

$$\sum_{n<v} M(n)^2 = -\sum_{h<v} h\, C_h^{[v]}, \qquad \sum_{h<v} h\, C_h^{[v]} = \sum_{\nu:\, x_{1,\nu}\le v} \sum_h h\, C_h^{(\nu)}. \tag{11.2}$$

Thus, in the h-weighted sum, the correlations between increments lying in different completed excursions cancel exactly; internal zero-run increments contribute nothing because $\mu(n) = 0$ there.

Proof. Apply Lemma 5.1 to the block $x_n = \mu(n)$, $1 \le n \le v$: then $S_t = M(t)$ and $S_v = M(v) = 0$, which gives the first identity. Because $M(v) = 0$, $\sum_{n\le v} M(n)^2 = \sum_{n<v} M(n)^2$. By (5.9), this equals $\sum_\nu Q_\nu$ over the completed excursions ending at or before v, and $Q_\nu = -\sum_h hC_h^{(\nu)}$ by (5.5). □

Theorem 11.3 (Weighted-correlation criterion)**.** [C] *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every zero } v \text{ of } M.$$

Proof. If RH holds, Theorem 8.6 and (11.2) give the bound. Conversely, fix a target $\varepsilon_0 > 0$ and choose $\varepsilon > 0$ so small that $(1+\varepsilon)(2+\varepsilon) \le 2+\varepsilon_0$. Let x be large and put $N = \lfloor x \rfloor$. If $M(N) = 0$, then $\sum_{n\le x} M(n)^2 = B_2(N)$ is already evaluated at a zero. Otherwise, let $u = \max\{n \le N : M(n) = 0\}$ and let $v > N$ be the next zero. Then $(u, v]$ is the completed excursion whose open prefix contains N, and Corollary 9.8 gives $v \le u^{1+\varepsilon} \le x^{1+\varepsilon}$ for all sufficiently large x. In either case, with $v = N$ in the zero case,

$$\sum_{n\le x} M(n)^2 \le \sum_{n<v} M(n)^2 \ll_\varepsilon v^{2+\varepsilon} \le x^{2+\varepsilon_0}.$$

Theorem 8.6 now applies. □

Proposition 11.4 (Square-root lag cutoff at zeros)**.** [C] *Let $v \to \infty$ through zeros of M, and let $K = K(v) \to \infty$ with $K \le v^{1/2}$. Then*

$$\sum_{h\le K} h\, C_h^{[v]} = o(K^2 v). \tag{11.3}$$

In particular,

$$\sum_{n<v} M(n)^2 = -\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} + o(v^2), \tag{11.4}$$

where the lower limit may be replaced by $\lfloor\sqrt{v}\rfloor$ without changing the statement.

Proof. By (11.1), $\sum_{h\le K} |C_h^{[v]}| = o(Kv)$. Therefore

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v).$$

Taking $K = \lfloor\sqrt{v}\rfloor$ and using Proposition 11.2 gives (11.4). □

Corollary 11.5 (RH after deleting all small lags)**.** [C] *RH holds if and only if, for every $\varepsilon > 0$,*

$$-\sum_{\sqrt{v}<h<v} h\, C_h^{[v]} \ll_\varepsilon v^{2+\varepsilon} \qquad \text{for every sufficiently large zero } v \text{ of } M.$$

Thus the lags $h \le \sqrt{v}$ are unconditionally negligible at the critical v^2 scale of the weighted-correlation criterion.

Theorem 11.6 (Zero-edge lag localization)**.** [N] *Let $v \to \infty$ through zeros of M. Then*

$$-\sum_{h<v} hC_h^{[v]} = \sum_{n<v} M(n)^2 = v^{1+2\Theta+o(1)}. \tag{11.5}$$

Moreover, for every fixed $0 \le \alpha < \Theta$,

$$\sum_{h\le v^\alpha} hC_h^{[v]} = o\left(\sum_{n<v} M(n)^2\right), \tag{11.6}$$

and hence

$$-\sum_{v^\alpha<h<v} hC_h^{[v]} = (1+o(1))\sum_{n<v} M(n)^2. \tag{11.7}$$

Thus the signed weighted correlation total is forced beyond every power v^α with $\alpha < \Theta$.

Proof. The first equality is Proposition 11.2, and the asymptotic exponent is Corollary 8.11. For $K = \lfloor v^\alpha \rfloor$ with $\alpha > 0$, (11.1) gives

$$\left|\sum_{h\le K} hC_h^{[v]}\right| \le K \sum_{h\le K} |C_h^{[v]}| = o(K^2 v) = o(v^{1+2\alpha}).$$

Choose a fixed δ_0 with $0 < \delta_0 < 2(\Theta-\alpha)$. Along zeros $v \to \infty$, Corollary 8.11 gives $B_2(v) \ge v^{1+2\Theta-\delta_0}$ for all sufficiently large v, while the small-lag contribution is $o(v^{1+2\alpha})$. Their ratio is therefore $o(v^{-2(\Theta-\alpha)+\delta_0}) = o(1)$. The case $\alpha = 0$ follows trivially from the fixed finite number of lags and $|C_h^{[v]}| \le v$. □

Remark 11.7 (Endpoint versus the universal square-root cutoff). Proposition 11.4 is uniform without knowing Θ and reaches the endpoint $K = \sqrt{v}$ at the absolute v^2 scale. Theorem 11.6 is stronger in relative terms but requires a strict power gap $\alpha < \Theta$; it does not settle the endpoint $\alpha = \Theta$. Under RH it says that every range $h \le v^{1/2-\delta}$ carries $o(1)$ of the total signed weighted correlation, while if RH fails the negligible range extends correspondingly beyond the square-root scale.

Remark 11.8 (Where the cancellation must occur). The trivial bound is $\sum_h h|C_h^{[v]}| \le \sum_h h(v-h) \le v^3/6$. RH requires the signed weighted sum to be $v^{2+o(1)}$. Absolute values cannot give this at the natural scale: even square-root cancellation $|C_h^{[v]}| \ll v^{1/2+o(1)}$ in every single lag would give only $\sum_h h|C_h^{[v]}| \ll v^{5/2+o(1)}$. Corollary 11.5 sharpens the location of the unresolved cancellation: at the v^2 scale, only lags $h > \sqrt{v}$ remain. By Proposition 11.2 the signed weighted identity itself decomposes excursion by excursion.

11.2. **Lag localization of the energy of an excursion.**

Proposition 11.9 (Energy beyond a lag cutoff)**.** [N-el] *For every excursion I_ν of length $\ell = \ell_\nu$ and every $1 \le h_0 < \ell$,*

$$\left|Q_\nu + \sum_{h_0<h<\ell} h\, C_h^{(\nu)}\right| \le h_0 \sum_{h\le h_0} \left|C_h^{(\nu)}\right|.$$

Consequently, if $\sum_{h\le h_0}|C_h^{(\nu)}|\le \delta h_0\ell$, *then*

$$\sum_{h_0<h<\ell} h\,C_h^{(\nu)} = -Q_\nu + O(\delta h_0^2\ell).$$

Proof. Split $Q_\nu = -\sum_{h<\ell} hC_h^{(\nu)}$ at h_0. □

Corollary 11.10 (Averaged Chowla on linear-scale intervals)**.** [C] *Let* $(u_j, u_j+L_j]$ *be intervals with* $L_j\to\infty$ *and* $u_j\le BL_j$ *for a fixed* B, *and let* $h_j\to\infty$ *with* $h_j = o(L_j)$. *Then*

$$\sum_{h\le h_j}\left|C_h^{(u_j,u_j+L_j]}\right| = o(h_jL_j).$$

Proof. Put $X_j = L_j - h_j$. Then $X_j\sim L_j$ and, for all large j, $u_j\le 2BX_j$. Apply the shifted form of [42, Theorem 1.6], whose hypotheses explicitly allow translation parameters $b_r\le AX_j$, to the two-point Möbius correlation. The same paper states immediately after Theorem 1.6 that the argument proving its averaged Chowla theorem also applies with λ replaced by μ. Taking one relative shift fixed and averaging the other over $1\le h\le h_j$, with the common translation $u_j = O(X_j)$ absorbed into the allowed b_r, gives

$$\sum_{h\le h_j}\left|\sum_{1\le n\le X_j}\mu(u_j+n)\mu(u_j+n+h)\right| = o(h_jX_j) = o(h_jL_j).$$

For each $h\le h_j$, extending the inner sum from $X_j = L_j - h_j$ to $L_j - h$ adds at most $h_j - h$ terms. Hence the total triangular boundary cost is

$$\sum_{h\le h_j}(h_j - h) = O(h_j^2) = o(h_jL_j),$$

which proves the claim. □

Corollary 11.11 (Large-lag concentration in linear-scale excursions)**.** [N] *Let* I_j *be excursions with* $\ell_j\to\infty$, $x_{0,j}\le B\ell_j$ *and heights* H_j *with* $H_j^3/\ell_j\to\infty$. *For every* $h_j\to\infty$ *with* $h_j^2\ell_j = o(H_j^3)$,

$$\sum_{h_j<h<\ell_j} h\,C_h^{(I_j)} = -(1+o(1))Q_{I_j}.$$

If in addition $\Theta > 1/2$, *there is a fixed* $\lambda' > 1$ *with* $x_{1,j}\le\lambda' x_{0,j}$, *and these excursions carry Kaczorowski–Pintz witnesses* $H_j\ge\frac12 x_{0,j}^{\Theta-\eta}$, *then for every fixed*

$$0<\rho<\frac{3(\Theta-\eta)-1}{2}$$

one may take $h_j = x_{0,j}^{\rho}$, *and hence*

$$\sum_{x_{0,j}^{\rho}<h<\ell_j} h\,C_h^{(I_j)} = -(1+o(1))Q_{I_j}.$$

Proof. By Corollary 11.10, $\sum_{h\le h_j}|C_h^{(I_j)}| = \delta_j h_j\ell_j$ with $\delta_j\to 0$. Proposition 11.9 gives an error at most $\delta_j h_j^2\ell_j = o(H_j^3) = o(Q_{I_j})$. In the witness case the hypotheses $x_{0,j}\le B\ell_j$ and $x_{1,j}\le\lambda' x_{0,j}$ give $\ell_j\asymp x_{0,j}$. Hence $H_j^3/\ell_j\gg x_{0,j}^{3\Theta-3\eta-1}$, so the displayed condition on ρ gives $h_j^2\ell_j = o(H_j^3)$. □

Remark 11.12 (Direction of the lag statement). Corollary 11.11 says that the energy lies beyond every cutoff h_j satisfying $h_j = o(H_j^{3/2}\ell_j^{-1/2})$; it does *not* say that the energy is concentrated only at lags larger than the endpoint scale $H_j^{3/2}\ell_j^{-1/2}$ itself.

11.3. **Uniform two-point cancellation in sublinear intervals.** The linear-scale restriction can be removed for mesoscopic lags by a Fourier-uniform estimate valid in every interval in the range displayed below. Theorem 1.5 of Matomäki–Teräväinen [44] gives the twisted Möbius estimate with the original logarithmic saving $A < 1/3$. Matomäki–Shao–Tao–Teräväinen record explicitly, in their discussion following equation (1.16), that [44, Remark 5.2] allows this range to be enlarged to $0 < A < 1$; see [43, discussion following (1.16)]. Thus, for every fixed $0 < A < 1$ and $\iota > 0$,

$$\sup_{\xi\in\mathbb{R}}\Big|\sum_{x<n\le x+L}\mu(n)e(\xi n)\Big| \ll_{A,\iota} \frac{L}{(\log x)^A} \qquad (x^{3/5+\iota}\le L\le x). \tag{11.8}$$

Theorem 11.13 (Uniform two-point Cesàro cancellation)**.** [C] *Fix $0 < A < 1$ and $\iota > 0$. Uniformly for all integers u, L, R with $u^{3/5+\iota} \le L \le u$ and $1 \le R \le L/2$,*

$$\frac{1}{R}\sum_{h=1}^{R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{A,\iota} \Big(\frac{L}{R}\Big)^{1/2}(\log u)^{-A}. \tag{11.9}$$

In particular, if $R \ge L(\log u)^{-2A+2\kappa}$ with $0 < \kappa < A$, the left side is $O((\log u)^{-\kappa})$.

Proof. Put $a_i = \mu(u+i)$ for $1 \le i \le L$ and $A_I(\xi) = \sum_i a_i e(i\xi)$. For $h > 0$ define $C_{-h}^{(u,u+L]} := \sum_{i=1}^{L-h} a_{i+h}a_i = C_h^{(u,u+L]}$ and put $C_0 = q = \sum_i a_i^2$. These are the Fourier coefficients of $|A_I(\xi)|^2$. Parseval gives the precise identity

$$\int_0^1 |A_I(\xi)|^4\,d\xi = q^2 + 2\sum_{h=1}^{L-1}|C_h^{(u,u+L]}|^2 \le \|A_I\|_\infty^2 q \le L\|A_I\|_\infty^2.$$

Hence Cauchy–Schwarz gives $\sum_{h\le R}|C_h| \le \sqrt{RL}\,\|A_I\|_\infty$, and since $L - h \ge L/2$ the left side of (11.9) is at most $2\|A_I\|_\infty/\sqrt{RL}$. Apply (11.8). □

Corollary 11.14 (Near-log-squared mesoscopic range)**.** [C] *Fix $\iota > 0$ and $0 < \delta < 1$. Uniformly for*

$$u^{3/5+\iota} \le L \le u, \qquad \frac{L}{(\log u)^{2-\delta}} \le R \le \frac{L}{2},$$

one has

$$\frac{1}{R}\sum_{h\le R}\frac{|C_h^{(u,u+L]}|}{L-h} \ll_{\iota,\delta} (\log u)^{-\delta/4}.$$

Proof. In Theorem 11.13 take $A = 1 - \delta/4$ and $\kappa = \delta/4$. □

Remark 11.15 (What is and is not new). Theorem 11.13 is a short Parseval–Cauchy–Schwarz corollary of the published Fourier estimate (11.8); it does not use the excursion condition. Its role here is to provide an arithmetic input on every adaptively selected interval, something that "almost all intervals" results cannot supply. The factor $(L/R)^{1/2}$ is inherent in this direct L^2-to-L^1 Cauchy–Schwarz step using only the displayed second-moment control of the C_h; additional information on the distribution of the correlations could in principle improve it.

Corollary 11.16 (Sublinear large-lag estimate)**.** [C] *Let $I = (u, u+L]$ be a complete excursion with $u^{3/5+\iota} \le L \le u$, and fix $0 < \delta < 1$. For all sufficiently large u, put*

$$R = \left\lceil \frac{L}{(\log u)^{2-\delta}} \right\rceil.$$

Then

$$\sum_{R<h<L} hC_h^{(I)} = -Q_I + O_{\iota,\delta}\Big(\frac{L^3}{(\log u)^{4-7\delta/4}}\Big). \tag{11.10}$$

Consequently, if $Q_I L^{-3}(\log u)^{4-7\delta/4} \to \infty$, *then the left side is* $-(1+o(1))Q_I$.

Proof. Corollary 11.14 gives $\sum_{h\le R}|C_h^{(I)}| \ll RL(\log u)^{-\delta/4}$. Hence

$$\left|\sum_{h\le R} hC_h^{(I)}\right| \le R\sum_{h\le R}|C_h^{(I)}| \ll R^2L(\log u)^{-\delta/4},$$

which is the error in (11.10) after inserting the chosen value of R into $Q_I = -\sum_{h<L} hC_h^{(I)}$. □

Remark 11.17 (Connection with wide peaks). The quantitative form of the short-interval estimate in [44, Theorem 1.1] gives, after the blocking argument of Lemma 9.31, $\omega_u \ll_\varepsilon (\log u)^{-1/3+\varepsilon} + u^{-c}$ for some $c > 0$ (depending on the two fixed short-interval exponents used there). Thus, in the regime $H/h_u \to \infty$ of Proposition 9.32,

$$Q_I \gg_\varepsilon H^3(\log u)^{1/3-\varepsilon}.$$

Combining this with Corollary 11.16, a sufficient condition for large-lag concentration is

$$\left(\frac{H}{L}\right)^3 (\log u)^{13/3-7\delta/4-\varepsilon} \longrightarrow \infty.$$

For linearly localized Kaczorowski–Pintz excursions satisfying $v \le \lambda u$ with a fixed $\lambda < 2$, this machinery applies when $\Theta > 3/5$ (after choosing the loss parameters so that $\Theta - \eta > 3/5 + \iota$); the remaining obstruction is that the excursion may be much longer than its height.

11.4. **Fejér reduction and the weaker local-variance target.**

Remark 11.18 (Excursion geometry alone does not give cancellation). For the tent excursion $a_i = +1$ $(i \le k)$, $a_i = -1$ $(k < i \le 2k)$, which satisfies the bridge and one-sign constraints of a primitive excursion, $C_h = 2k - 3h$ for $1 \le h < k$. Hence for $R \to \infty$, $R = o(k)$,

$$\frac{1}{R}\sum_{h\le R}\frac{|C_h|}{2k-h} \longrightarrow 1, \qquad \sum_{h<R}(R-h)C_h \asymp kR^2,$$

whereas $RL \asymp kR$. Arithmetic input is indispensable.

Lemma 11.19 (Finite Fejér identity)**.** [L] *For a real sequence* $(a_i)_{1\le i\le L}$, *extended by* 0 *outside* $[1, L]$, *put*

$$q = \sum_{i=1}^{L} a_i^2, \qquad C_h = \sum_{i=1}^{L-h} a_i a_{i+h}, \qquad F_R := \sum_{h=1}^{R-1}(R-h)C_h.$$

Then for $1 \le R \le L$,

$$Rq + 2F_R = \sum_{m\in\mathbb{Z}}\left(\sum_{j=1}^{R} a_{m+j}\right)^2. \tag{11.11}$$

Proof. Expand the square. Each diagonal term occurs R times, and a pair at lag h, $1 \le h < R$, occurs $2(R-h)$ times. □

Proposition 11.20 (Bridge–Fejér identity and interior variance)**.** [N-el] *Let* $I = (u, u+L]$ *be a complete excursion, put* $a_i = \mu(u+i)$ *and* $S_t = \sum_{i=1}^{t} a_i = M(u+t)$ *for* $0 \le t \le L$. *Then* $S_0 = S_L = 0$, *and for every* $1 \le R < L$,

$$Rq + 2F_R = 2Q_I - 2\sum_{t=1}^{L-1-R} S_t S_{t+R}. \tag{11.12}$$

Moreover

$$Rq + 2F_R = E_{\mathrm{bd}}(R) + \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2, \tag{11.13}$$

where

$$E_{\mathrm{bd}}(R) = \sum_{t=1}^{R-1} S_t^2 + \sum_{t=L-R+1}^{L-1} S_t^2 = O(R^3).$$

Consequently, if $R = o(\sqrt{L})$*, then*

$$F_R = o(RL) \quad \Longleftrightarrow \quad \sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL). \tag{11.14}$$

If in addition $L \geq u^{1/2+\varepsilon}$ *and* $L \leq u$*, then*

$$q = \left(\frac{6}{\pi^2} + o(1)\right) L, \tag{11.15}$$

and the right side of (11.14) *is equivalently*

$$\sum_{m=0}^{L-R} \big(M(u+m+R) - M(u+m)\big)^2 = \left(\frac{6}{\pi^2} + o(1)\right) RL. \tag{11.16}$$

Proof. Extend S_t by 0 outside $0 \leq t \leq L$. Since $\sum_{j=1}^{R} a_{m+j} = S_{m+R} - S_m$, (11.11) gives

$$Rq + 2F_R = \sum_m (S_{m+R} - S_m)^2 = 2\sum_m S_m^2 - 2\sum_m S_m S_{m+R},$$

The natural range of the last sum is $0 \leq m \leq L-R$, but the terms $m = 0$ and $m = L-R$ vanish because $S_0 = S_L = 0$, leaving $1 \leq t \leq L-1-R$; this is (11.12). Splitting the same sum into $m < 0$, $0 \leq m \leq L-R$, and $m > L-R$ gives (11.13): for $-R < m < 0$ the contribution is S_{m+R}^2, for $L-R < m < L$ it is S_m^2, and the endpoint terms vanish because $S_0 = S_L = 0$. Since $|a_i| \leq 1$ and the path returns to zero, $|S_t| \leq t$ and $|S_{L-t}| \leq t$, whence $E_{\mathrm{bd}}(R) \leq 2\sum_{t<R} t^2 = O(R^3)$. Thus $E_{\mathrm{bd}}(R) = o(RL)$ when $R = o(\sqrt{L})$, proving the equivalence. Finally the classical estimate $\sum_{n \leq x} \mu(n)^2 = (6/\pi^2)x + O(\sqrt{x})$ gives $q = (6/\pi^2)L + O(\sqrt{u+L})$. Under $u^{1/2+\varepsilon} \leq L \leq u$,

$$\frac{\sqrt{u+L}}{L} \leq \frac{\sqrt{2u}}{u^{1/2+\varepsilon}} \ll u^{-\varepsilon},$$

so the error is uniform and (11.15) follows. □

Remark 11.21 (Fejér density at frequency zero). With $A_I(\xi) = \sum_{i=1}^{L} a_i e(i\xi)$ and

$$K_R(\xi) = \frac{1}{R}\left|\sum_{j=1}^{R} e(j\xi)\right|^2,$$

one has

$$q + \frac{2F_R}{R} = \int_0^1 |A_I(\xi)|^2 K_R(\xi)\, d\xi.$$

Thus $F_R = o(RL)$ says that the Fejér average of the local spectral density at frequency 0, at resolution $1/R$, equals $q + o(L)$. It is not a statement that $|A_I(\xi)|^2$ is spectrally flat on the whole circle.

Remark 11.22 (Three distinct targets). The following conditions must not be conflated.

First, absolute averaged Chowla, $\sum_{h<R}|C_h| = o(RL)$, implies only $|F_R| \le R\sum_{h<R}|C_h| = o(R^2L)$, not the sharper $F_R = o(RL)$ when $R \to \infty$. Conversely, $F_R = o(RL)$ allows cancellation between the C_h and does not control their absolute values.

Second, F_R does not by itself control the small-lag contribution to the excursion energy. If

$$V_R := \sum_{h<R} hC_h,$$

then exactly

$$F_R + V_R = R\sum_{h<R} C_h. \tag{11.17}$$

Thus a Fejér estimate alone does not imply $V_R = o(Q_I)$.

Third, Proposition 11.20 requires both endpoints of the block to be zeros of M. For a block beginning at a zero but ending before the return, additional boundary terms remain.

Remark 11.23 (Why the variance target may go below 3/5). For the Fejér problem one does not need the stronger absolute estimate $\sum|C_h|$. By Proposition 11.20, on a complete excursion and for $R = o(\sqrt{L})$ it is enough to prove the diagonal local-variance law

$$\sum_{m=0}^{L-R}\big(M(u+m+R) - M(u+m)\big)^2 = Rq + o(RL).$$

This is strictly weaker than absolute averaged Chowla and is therefore a plausible target below the Fourier-uniform threshold $L \ge u^{3/5+\iota}$. The all-interval U^2/Fourier estimates currently used above do not, by themselves, supply this asymptotic for every adaptively selected block in the range $L \ge u^{1/2+\varepsilon}$. Results holding for almost all starting points cannot be substituted, because excursion endpoints are selected by M itself.

No fixed-prefix formula for the increments in (11.16) is asserted here. In particular, the identity $\sum_{d\le x}\mu(d)\lfloor x/d\rfloor = 1$ introduces lower-scale terms such as $M(x/2)$ when one truncates at $x/2$; omitting those terms would give an invalid reduction.

12. Extreme values in stochastic models of excursions

The counts of Section 9.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\mathfrak{X}, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \le x_{0,k+1}$, and real random variables Z_k (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_\flat$; let $k_{\mathcal{F}}(1) < k_{\mathcal{F}}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \le X\} \qquad (\max\emptyset = -\infty).$$

The lemmas of Section 9.1 hold for every realization.

Hypothesis 12.1 (SG_ϖ)**.** There are $C, \kappa_0, \varpi > 0$ such that for every $j \ge 1$ and $z \ge 0$

$$\Pr\big(k_{\mathcal{F}}(j) \text{ exists and } Z_{k_{\mathcal{F}}(j)} > z\big) \le Ce^{-\kappa_0 z^\varpi}.$$

For $\varpi = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

Theorem 12.2 (Extreme scale from a count)**.** [N-el] *Assume* (SG_ϖ) *for* $\mathcal{F}$ *and suppose that* $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ *for every realization, with* $\mathcal{N}(X) \ge 1$ *deterministic. Then*

$$\Pr\big(Z_{\max,\mathcal{F}}(X) > z\big) \le \mathcal{N}(X)\, Ce^{-\kappa_0 z^\varpi}. \tag{12.1}$$

Consequently, for every $\delta > 0$,

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{\kappa_0}\log\mathcal{N}(X)\Big)^{1/\varpi}\Big) \le C\,\mathcal{N}(X)^{-\delta}, \tag{12.2}$$

and $Z_{\max,\mathcal{F}}(X) = O_{\Pr}\big((\log\mathcal{N}(X))^{1/\varpi}\big)$ as $\mathcal{N}(X)\to\infty$.

Proof. The members of $\mathcal{F}$ starting at most X are the first $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{1\le j\le\lfloor\mathcal{N}(X)\rfloor}\{k_{\mathcal{F}}(j) \text{ exists},\ Z_{k_{\mathcal{F}}(j)} > z\}$, and (12.1) is the union bound. Substituting $\kappa_0 z^{\varpi} = (1+\delta)\log\mathcal{N}(X)$ gives (12.2). □

Corollary 12.3 (Triple-logarithmic scale on the square family)**.** [N-el] *Assume (SG) for the square-exceptional family $\mathcal{F}_{x^2}$ of the model. Then, for $X \ge e^e$ and every $\delta > 0$,*

$$\Pr\big(Z_{\max,\square}(X) > z\big) \le 4C(\log\log X)e^{-\kappa_0 z^2}, \tag{12.3}$$

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{\kappa_0}\log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{12.4}$$

Thus $Z_{\max,\square}(X) = O_{\Pr}(\sqrt{\log\log\log X})$. The same holds for the family $\mathcal{F}_{cx^2}$ of Lemma 9.2 with a constant depending on c.

Proof. Use (9.2) in (12.1) and note that $e^{-\kappa_0 z^2} = (\log\log X)^{-(1+\delta)}$ at the stated z. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to $\mathcal{N}$ gives an almost sure bound with the same constant.

Theorem 12.4 (Almost sure extreme scale)**.** [N-el] *Assume* (SG$_{\varpi}$) *for $\mathcal{F}$ and $E_{\mathcal{F}}(X) \le \mathcal{N}(X)$ for every realization, where $\mathcal{N}$ is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X\to\infty}\frac{Z_{\max,\mathcal{F}}(X)}{(\log\mathcal{N}(X))^{1/\varpi}} \le \kappa_0^{-1/\varpi}.$$

Proof. Fix $\delta > 0$. For $j \ge j_0$ let $X_j := \inf\{X : \mathcal{N}(X) \ge 2^j\}$. By continuity, monotonicity and unboundedness, after increasing j_0 if necessary one has $\mathcal{N}(X_j) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/\kappa_0\big)^{1/\varpi}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (12.1), $\Pr(A_j) \le 2^{j+1}Ce^{-(1+\delta)(j+1)\log 2} = C2^{-(j+1)\delta}$, which is summable, so almost surely A_j occurs only finitely often. For $X_j \le X \le X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in X gives $Z_{\max,\mathcal{F}}(X) \le z_j$ outside A_j, while $\log\mathcal{N}(X) \ge j\log 2$. Hence the lim sup is at most $\lim_j\big((1+\delta)(j+1)/(\kappa_0 j)\big)^{1/\varpi} = ((1+\delta)/\kappa_0)^{1/\varpi}$. Let $\delta\downarrow 0$ along a countable sequence. □

With $\mathcal{N}_{\square}(X) = 2 + \log_2(\log X/\log 2)$, which is continuous and increasing, Theorem 12.4 gives under (SG)

$$\limsup_{X\to\infty}\frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \le \kappa_0^{-1/2} \quad \text{almost surely.}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most $X+1$; Lemma 9.4; Lemma 9.1), Theorems 12.2 and 12.4 give three levels:

family	members starting $\le X$	extreme scale under (SG)
all excursions	$\le X+1$	$\sqrt{\log X}$
$x_1 > \lambda x_0$	$O(\log X)$	$\sqrt{\log\log X}$
$x_1 > cx_0^2$ (or x_0^A)	$O(\log\log X)$	$\sqrt{\log\log\log X}$

Each level of lacunarity removes one logarithm from the extreme scale. Under (SG$_{\varpi}$) the last column reads $(\log X)^{1/\varpi}$, $(\log\log X)^{1/\varpi}$, $(\log\log\log X)^{1/\varpi}$.

Remark 12.5 (Scope). For M itself the square-exceptional family is finite by Corollary 9.8, so Theorems 12.2–12.4 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 9.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

13. A one-frequency model for the black curve

Heuristic 13.1 (Scale and period)**.** Motivated by (6.1), model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, and retain only the lowest zero γ_1 (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so the black curve oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{13.1}$$

the scale of Corollary 7.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in x. In this model consecutive zeros of M are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$, the threshold of Corollary 9.13(iii): a model excursion is exactly λ_*-long.

14. Numerical data

The following census covers $n \le N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions, recording $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{14.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\mathfrak{b}_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{14.2}$$

Since $|M(x)| \le 0.571\sqrt{x}$ for $33 \le x \le 10^{16}$ [37], and $|M(n)| \le \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $C_0 = 1$ is admissible in Lemma 7.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \le \frac{4}{5}N^{5/4} = 5.317 \times 10^{10}$ against the observed 1.3433×10^{10}, a ratio of 3.96. Heuristic 8.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants. Reproducing these values requires the excursion file, the area-integration convention of Proposition 4.2, and the terminal convention of Definition 3.6.

15. Conclusions

What is new and what is reformulation. New: the excursion skeleton and its exact geometry; the counting lemmas for disjoint intervals; the horizon-localization theorem with threshold $\lambda_* = e^{\pi/\gamma_1}$ and the descending horizons; the localization of returns $x_1 \le x_0^{1+o(1)}$; the exact area exponent $1 + \Theta/2$, which settles the equivalence of the $x^{5/4+\varepsilon}$ area bound with RH; the two-sided localization of large values and the mean logarithmic length $\le \pi/\gamma_1$ of witness excursions; wide peaks; the prefix-dependency cutoff $\chi_* = \lambda_*/2$; channel descent with quantified persistence; the weighted correlation identity at zeros with Θ-edge lag localization; the large-prime decomposition $M = M_U^{\mathrm{sm}} + M_U^{\mathrm{lp}}$ with both parts of size $t/\log^2 t$. Reformulations: the L^1, L^2, energy and correlation forms of Littlewood's criterion, the exact exponents of $B_1, B_2, \mathcal{E}$ from Pintz, the excursion-block series of $1/\zeta$, and the transport of averaged Chowla, Fourier uniformity and short-interval cancellation to excursion blocks.

Lines this paper can help to pursue.

(1) *Lowering λ_*.* By Corollary 9.29 the witness excursions have mean logarithmic length $\le \pi/\gamma_1$; any positional information giving a fixed surplus for the witnesses beyond λx_0 lowers the horizon below λ_* (Remark 9.30).
(2) *Local variance below 3/5.* On complete excursions with $L \ge u^{1/2+\varepsilon}$ the diagonal law $\sum_m (M(u+m+R) - M(u+m))^2 = (6/\pi^2 + o(1))RL$ for some $R \to \infty$, $R = o(\sqrt{L})$, is equivalent to $F_R = o(RL)$ (Proposition 11.20) and is strictly weaker than absolute averaged Chowla; it is the natural target for adaptively selected blocks below the Fourier-uniform range.
(3) *Absolute two-point cancellation on adaptive blocks* of length $x_0^{1/2+\varepsilon} \le \ell < x_0^{3/5}$, extending Theorem 11.13.
(4) *The lag endpoint $\alpha = \Theta$* in Theorem 11.6: under RH, relative negligibility of $h \le v^{1/2}$ would locate the whole weighted correlation mass at lags $h > v^{1/2+o(1)}$.
(5) *Collective use of persistence.* Channel descent iterates with a constant loss; the multi-index persistence of Lemma 10.8 and cancellation inside the deep remainder $\mathcal{R}_K$ are the two places where the loss can be recovered.
(6) *Layer depth.* The depth $k(u)$ of Möbius layers needed to force the RH scale on the localized window is a quantitative invariant; fixed k is excluded (Corollary 10.29), and estimates uniform in $k \asymp \log\log u$ are the next step.
(7) *Exponent meters.* B_1, B_2, $\mathcal{E}$, $\mathcal{A}_{\mathrm{abs}}$ and the block series all have exponents that are explicit affine functions of Θ; they give several independent observables of Θ with different weights on long and short excursions.
(8) *Boundary of the block series* at $\Re s = \Theta$, and a third bridge relation for complete excursions involving the defect $\ell_\nu/2 - H_\nu$.
(9) *Stochastic models.* Tail hypotheses of type (SG) for $H/\sqrt{\ell}$ in random-walk models, where the extreme-value theorems of Section 12 apply.

Lines shown here not to work.

(1) Unweighted signed lag averages inside a bridge are $O(\ell^{-1})$ for algebraic reasons (Remark 11.1); only absolute or weighted averages carry arithmetic content.
(2) Square-root cancellation in each individual lag gives only $v^{5/2+o(1)}$ for the weighted sum, not $v^{2+o(1)}$ (Remark 11.8).
(3) Pointwise channel descent loses a factor $k^\vartheta/(2K) < 1/2$ per step and cannot be iterated to a bound (Remark 10.11).
(4) Termwise absolute-value contraction of the kernel gives nothing beyond $|M(u+r)| \le r$ (Proposition 10.23).
(5) The relaxed divisor-sum class that keeps only future-window constraints admits ballistic paths (Proposition 10.19).
(6) Any fixed number of multiplicative layers determines only $o(u)$ increments, whose sum is $\asymp u(\log\log u)^k/\log u$ (Corollary 10.29).
(7) Dependency reduction alone: the window is determined by the prefix up to $\chi_* u$, but determinacy is not cancellation (Remark 10.35).
(8) The global Catalan maximum tail is the wrong statistic at square horizons (Remark 9.18), and Catalan selector bounds are one-sided (Remark 9.22).
(9) Short-interval cancellation $|M(x+h)-M(x)| = o(h)$ and energy bounds do not force returns and cannot lower λ_* (Remark 9.30).
(10) Power and fractional-square horizons are inactive for M (Corollary 9.8); only the horizons $x_0\psi(x_0)$ and λx_0 carry information for the actual Mertens excursions.
(11) The truncated bridge identity cannot be applied inside a horizon window (Remark 10.21).

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